

Refraction at a spherical surface

① for Convex Spherical surface -

In $\triangle ONC$

$$i = \alpha + \beta \quad \text{--- (1) (external Angle property)}$$

In $\triangle NCI$

$$\beta = r + \gamma \quad \text{(external angle property)}$$

$$\Rightarrow r = \beta - \gamma \quad \text{--- (2)}$$

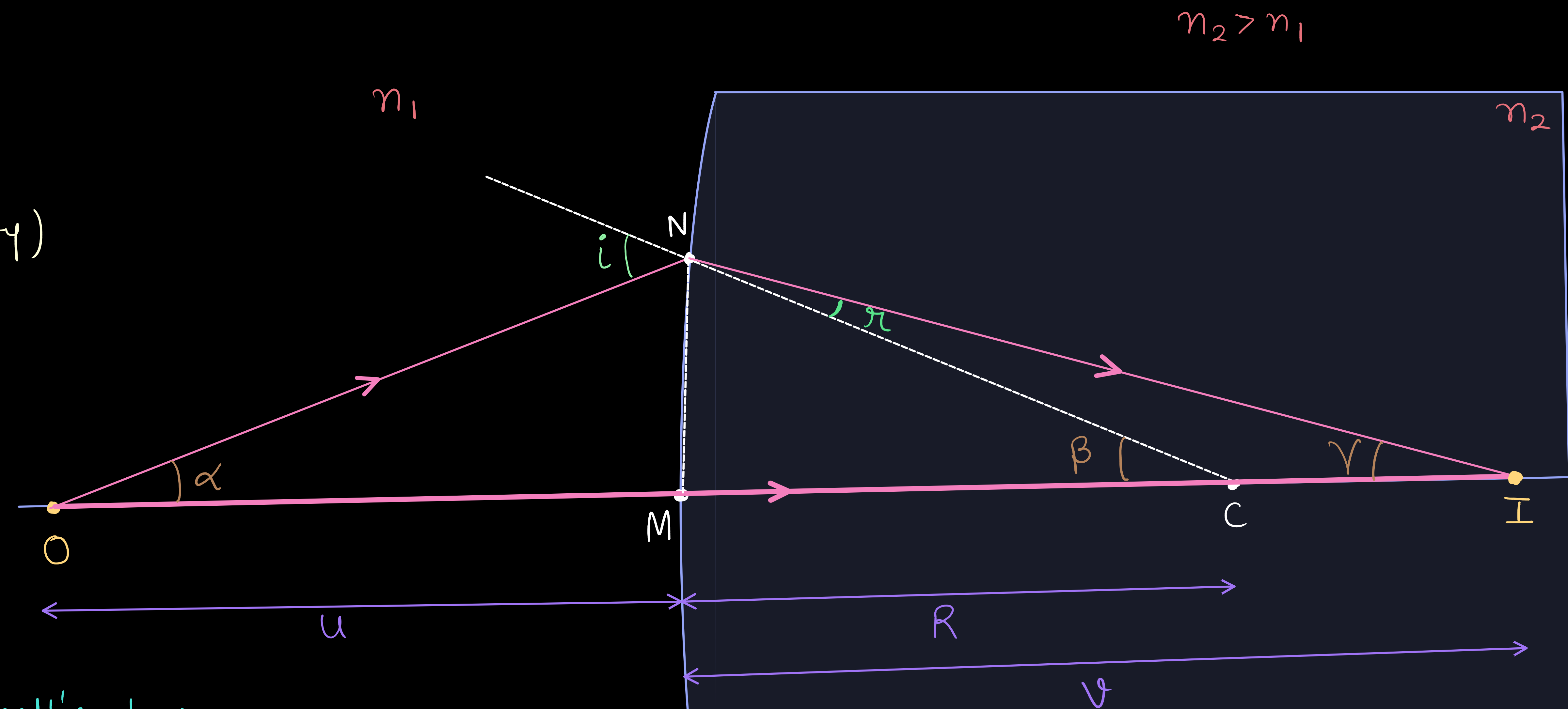
for paraxial rays -

i, r, α, β and γ are small

$$\therefore \tan \alpha \approx \alpha = \frac{NM}{OM} \quad \text{--- (3)}$$

$$\tan \beta \approx \beta = \frac{NM}{MC} \quad \text{--- (4)}$$

$$\tan \gamma \approx \gamma = \frac{NM}{MI} \quad \text{--- (5)}$$



Now, using Snell's law -

$$n_1 \sin i = n_2 \sin r$$

$$n_1 i = n_2 r \quad \text{--- (6) } (\because \sin i \approx i, \sin r \approx r)$$

$$n_1 (\alpha + \beta) = n_2 (\beta - \gamma)$$

Substitute values of α, β and γ from eqⁿ (3), (4) & (5)

$$n_1 \left(\frac{NM}{OM} + \frac{NM}{MC} \right) = n_2 \left(\frac{NM}{MC} - \frac{NM}{MI} \right)$$

Refraction at a spherical surface

$$n_1 \left(\frac{\cancel{NM}}{OM} + \frac{\cancel{NM}}{MC} \right) = n_2 \left(\frac{\cancel{NM}}{MC} - \frac{\cancel{NM}}{MI} \right)$$

$$n_1 \left(\frac{1}{OM} + \frac{1}{MC} \right) = n_2 \left(\frac{1}{MC} - \frac{1}{MI} \right)$$

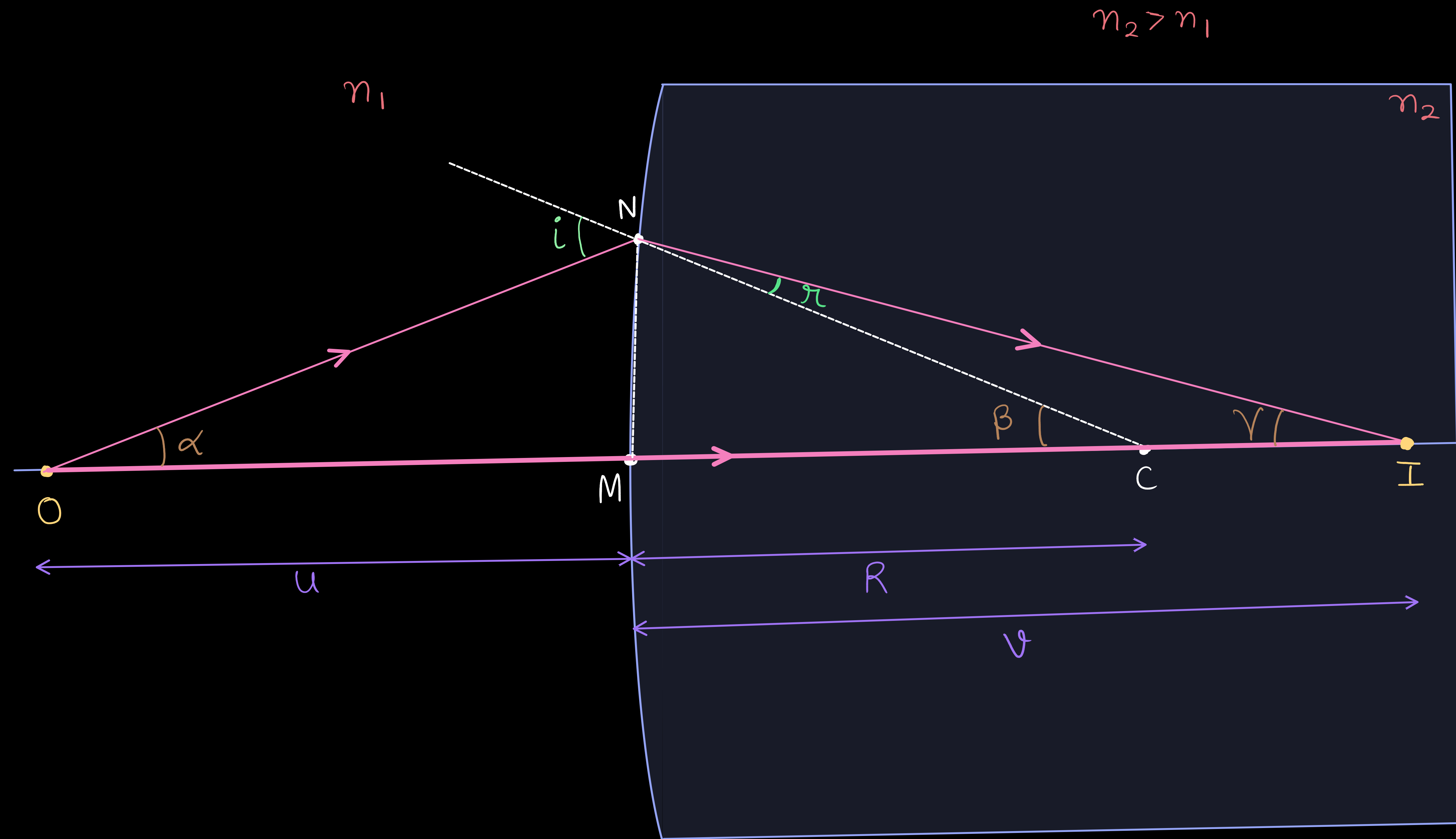
$$\frac{n_1}{OM} + \frac{n_1}{MC} = \frac{n_2}{MC} - \frac{n_2}{MI}$$

$$\frac{n_1}{OM} + \frac{n_2}{MI} = \frac{(n_2 - n_1)}{MC}$$

Now, using sign convention -

$$OM = -u, \quad MI = v, \quad MC = R$$

$$\boxed{-\frac{n_1}{u} + \frac{n_2}{v} = \frac{(n_2 - n_1)}{R}} \quad \star \star \star$$



Refraction at a spherical surface

② for convex surface (Virtual image)

In $\triangle NOC$

$$i = \beta + \gamma \quad \text{--- (1) (external angle)}$$

In $\triangle NIC$

$$r = \alpha + \gamma \quad \text{--- (2) (external angle)}$$

for paraxial rays -

$$\tan \alpha \approx \alpha = \frac{NM}{MI} \quad \text{--- (3)}$$

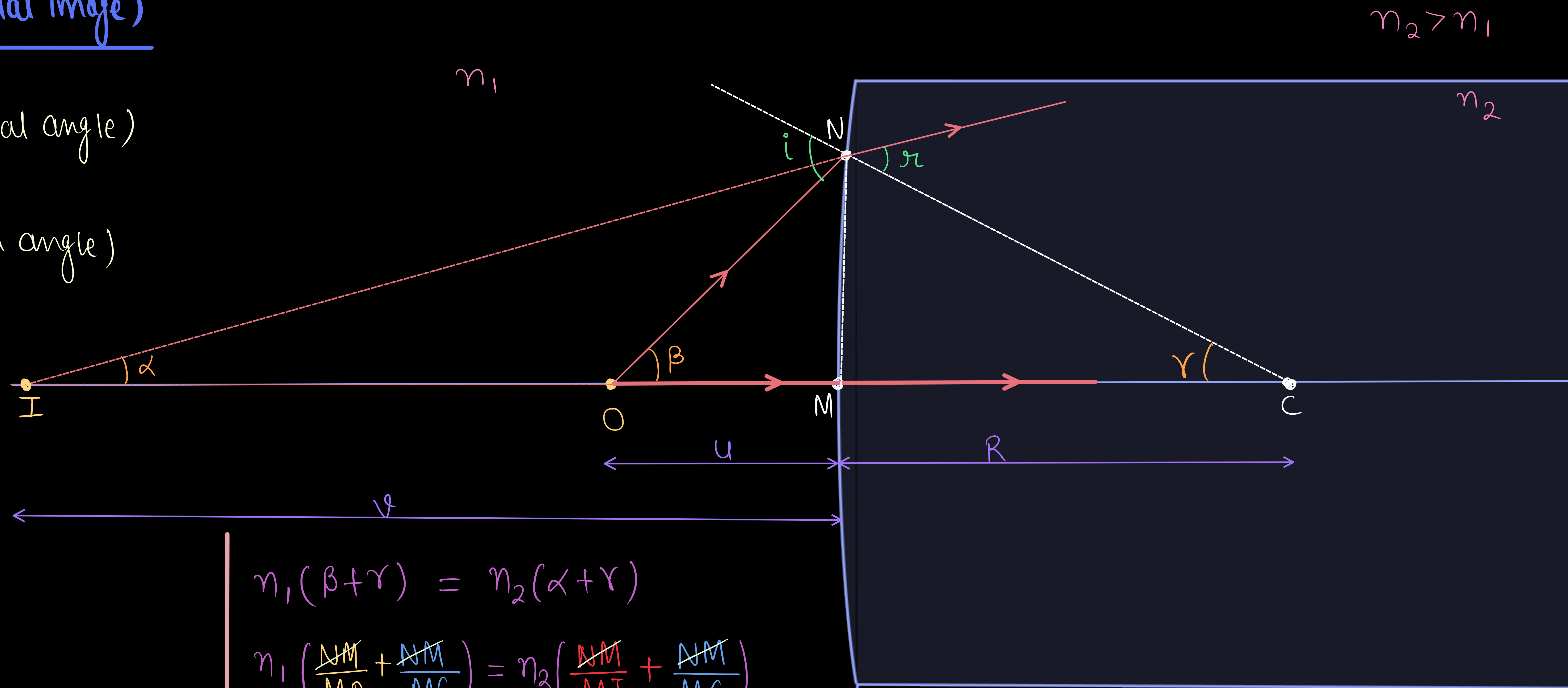
$$\tan \beta \approx \beta = \frac{NM}{MO} \quad \text{--- (4)}$$

$$\tan \gamma \approx \gamma = \frac{NM}{MC} \quad \text{--- (5)}$$

Now, using Snell's law -

$$n_1 \sin i = n_2 \sin r$$

$$\text{or } n_1 i = n_2 r \quad (\because \sin i \approx i, \sin r \approx r)$$



$$n_1(\beta + \gamma) = n_2(\alpha + \gamma)$$

$$n_1 \left(\frac{NM}{MO} + \frac{NM}{MC} \right) = n_2 \left(\frac{NM}{MI} + \frac{NM}{MC} \right)$$

$$n_1 \left(\frac{1}{MO} + \frac{1}{MC} \right) = n_2 \left(\frac{1}{MI} + \frac{1}{MC} \right)$$

$$\frac{n_1}{MO} + \frac{n_1}{MC} = \frac{n_2}{MI} + \frac{n_2}{MC}$$

$$\frac{n_1}{MO} - \frac{n_2}{MI} = \frac{n_2 - n_1}{MC} \quad \text{--- (6)}$$

using sign convention $\rightarrow MO = -u, MI = -v, MC = R$

$$\boxed{-\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}}$$

Refraction at a spherical surface

③ for convex surface (Object is in denser Medium)

In $\triangle NCO$ -

$$\beta = i + \gamma \text{ (external angle)}$$

$$\text{or } i = \beta - \gamma \text{ --- (1)}$$

In $\triangle NIC$ -

$$r = \alpha + \beta \text{ (ext. angle)}$$

for paraxial rays -

$$\sin \alpha \approx \alpha = \frac{NM}{MI} \text{ --- (3)}$$

$$\sin \beta \approx \beta = \frac{NM}{MC} \text{ --- (4)}$$

$$\sin \gamma \approx \gamma = \frac{NM}{MO} \text{ --- (5)}$$

using snell's law -

$$n_2 \sin i = n_1 \sin r$$

$$n_2 i = n_1 r$$

$$n_2 (\beta - \gamma) = n_1 (\alpha + \beta)$$

$$n_2 \left(\frac{NM}{MC} - \frac{NM}{MO} \right) = n_1 \left(\frac{NM}{MI} + \frac{NM}{MC} \right)$$

$$n_2 \left(\frac{1}{MC} - \frac{1}{MO} \right) = n_1 \left(\frac{1}{MI} + \frac{1}{MC} \right)$$

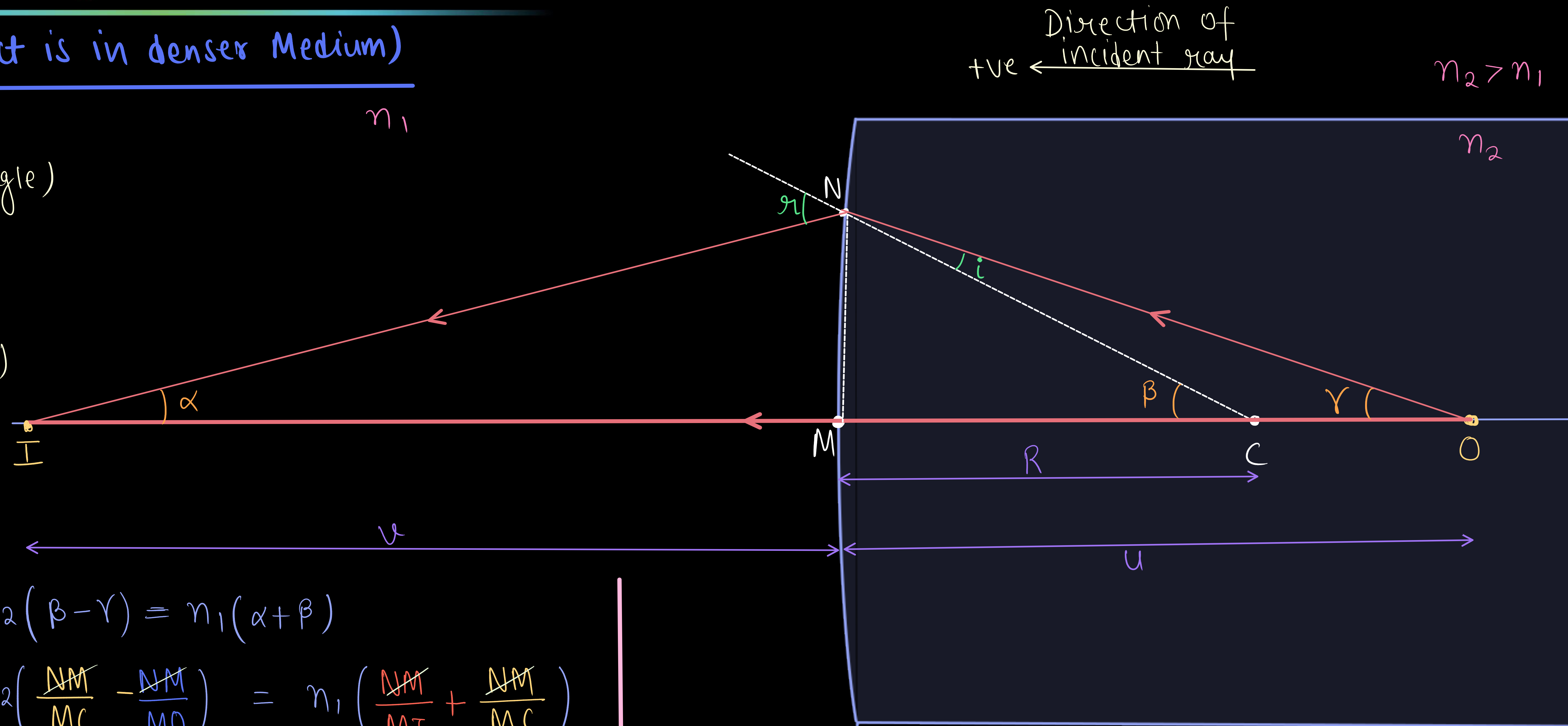
$$\frac{n_2}{MC} - \frac{n_2}{MO} = \frac{n_1}{MI} + \frac{n_1}{MC}$$

$$\frac{n_2 - n_1}{MC} = \frac{n_1}{MI} + \frac{n_2}{MO}$$

using sign convention - $MC = -R$, $MO = -u$, $MI = v$

$$\frac{n_2 - n_1}{-R} = \frac{n_1}{v} - \frac{n_2}{u} \text{ or}$$

$$\boxed{\frac{n_1}{v} - \frac{n_2}{u} = \frac{n_1 - n_2}{R}}$$



Refraction at a spherical surface

④ for concave spherical surface -

$$n_2 > n_1$$

In ΔNCO

$$\gamma = \alpha + i \text{ (ext. Ang.)}$$

$$\text{or } i = \gamma - \alpha \text{ --- (1)}$$

In ΔNCI

$$\gamma = \beta + r \text{ (ext. Ang.)}$$

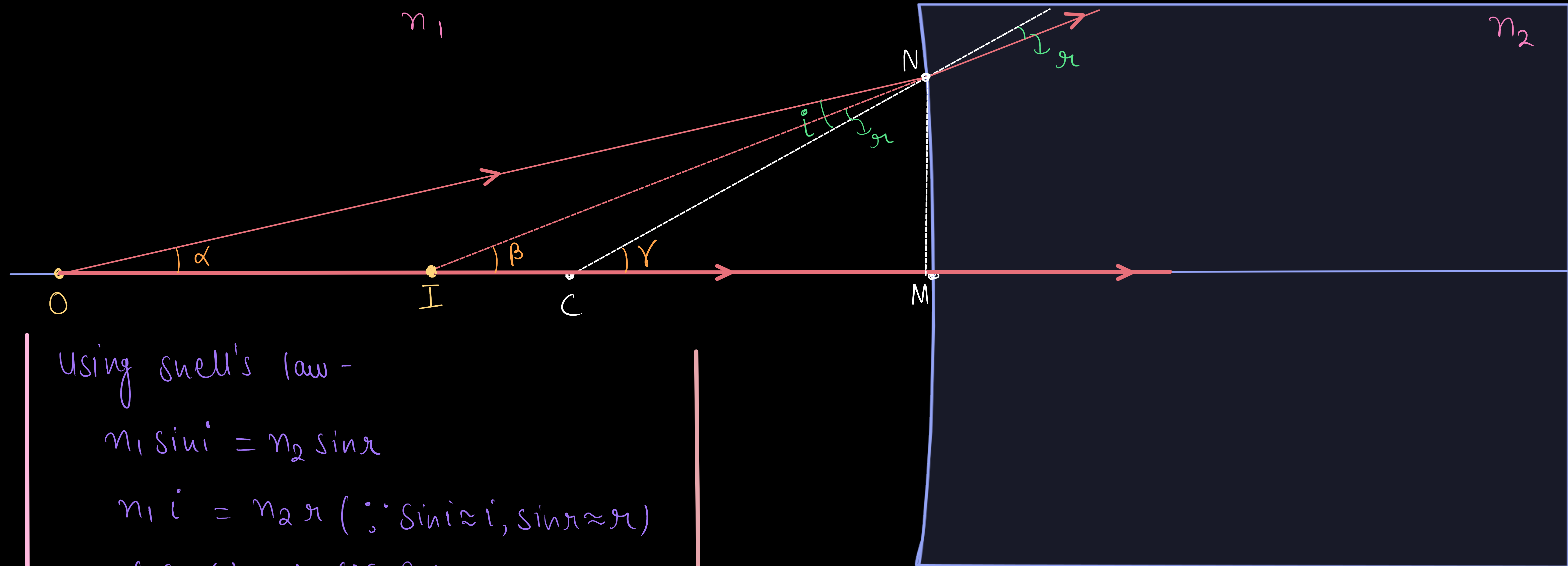
$$\text{or } r = \gamma - \beta \text{ --- (2)}$$

for paraxial rays -

$$\tan \alpha \approx \alpha = \frac{NM}{MO} \text{ --- (3)}$$

$$\tan \beta \approx \beta = \frac{NM}{MI} \text{ --- (3)}$$

$$\tan \gamma \approx \gamma = \frac{NM}{MC} \text{ --- (4)}$$



using snell's law -

$$n_1 \sin i = n_2 \sin r$$

$$n_1 i = n_2 r \quad (\because \sin i \approx i, \sin r \approx r)$$

$$n_1 (\gamma - \alpha) = n_2 (\gamma - \beta)$$

$$n_1 \left(\frac{NM}{MC} - \frac{NM}{MO} \right) = n_2 \left(\frac{NM}{MC} - \frac{NM}{MI} \right)$$

$$\frac{n_1}{MC} - \frac{n_1}{MO} = \frac{n_2}{MC} - \frac{n_2}{MI}$$

$$\frac{n_2}{MI} - \frac{n_1}{MO} = \frac{n_2 - n_1}{MC}$$

using sign convention, $MI = -v$, $MO = -u$, $MC = -R$

$$\frac{n_2}{-v} - \frac{n_1}{-u} = \frac{n_2 - n_1}{-R} \text{ or}$$

$$\boxed{\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}}$$

Refraction at a spherical surface

⑤ for concave surface (object in denser Medium)

In ΔNCO -

$$i = \alpha + \gamma \quad \text{--- (1) (ext. Angle)}$$

In ΔNCI -

$$r = \alpha + \beta \quad \text{--- (2) (ext. Angle)}$$

For paraxial rays -

$$\tan \alpha \approx \alpha = \frac{NM}{MC} \quad \text{--- (3)}$$

$$\tan \beta \approx \beta = \frac{NM}{MI} \quad \text{--- (4)}$$

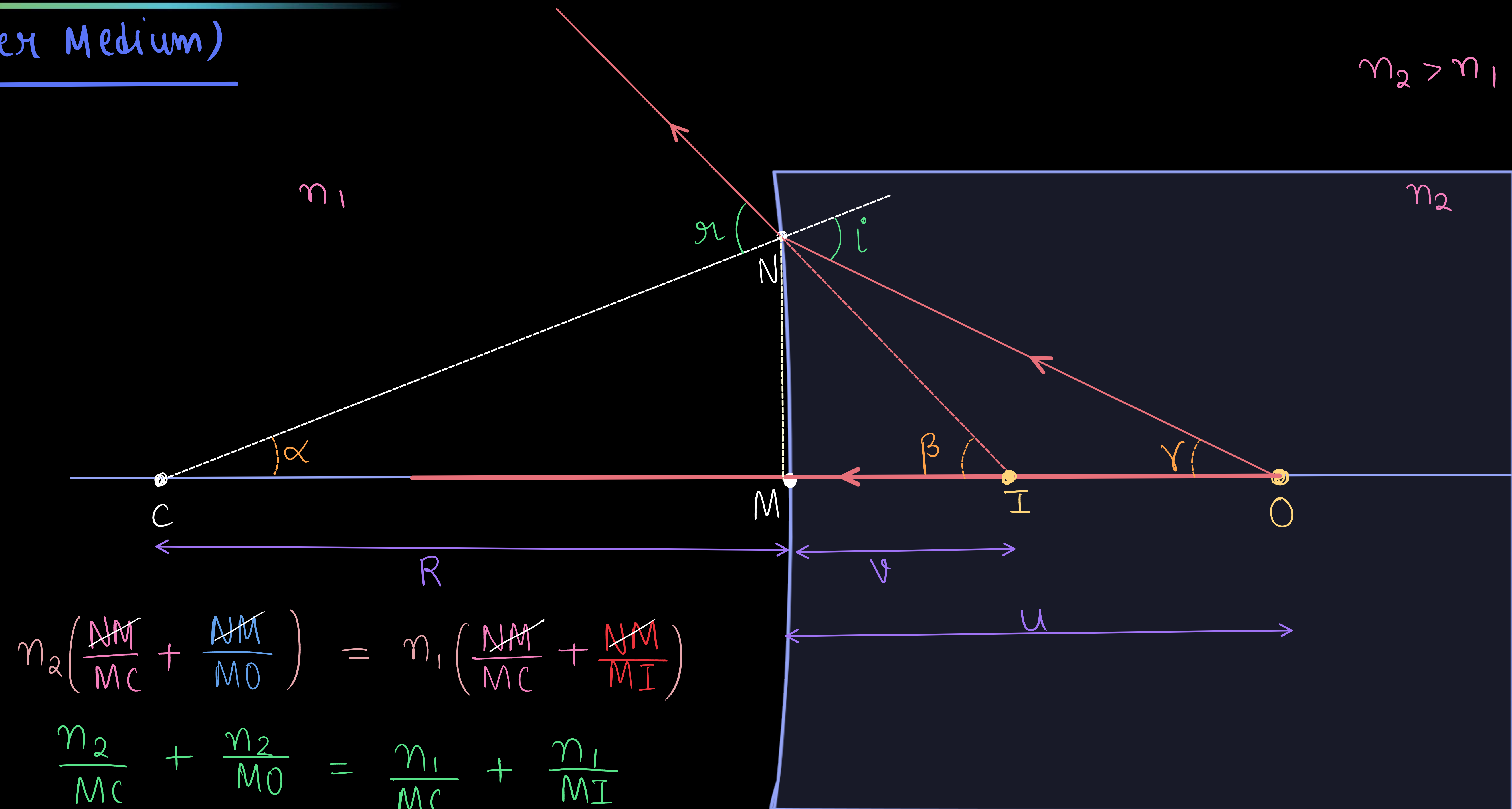
$$\tan \gamma \approx \gamma = \frac{NM}{MO} \quad \text{--- (5)}$$

using Snell's law -

$$n_2 \sin i = n_1 \sin r$$

$$\text{or } n_2 i = n_1 r \quad (\because \sin i \approx i, \sin r \approx r)$$

$$n_2(\alpha + \gamma) = n_1(\alpha + \beta)$$



$$n_2 \left(\frac{NM}{MC} + \frac{NM}{MO} \right) = n_1 \left(\frac{NM}{MC} + \frac{NM}{MI} \right)$$

$$\frac{n_2}{MC} + \frac{n_2}{MO} = \frac{n_1}{MC} + \frac{n_1}{MI}$$

$$\frac{n_2}{MO} - \frac{n_1}{MI} = \frac{n_1 - n_2}{MC}$$

using sign convention -

$$MO = -u, MI = -v, MC = R$$

$$\frac{n_2}{-u} - \frac{n_1}{-v} = \frac{n_1 - n_2}{R}$$

$$\boxed{\frac{n_1}{v} - \frac{n_2}{u} = \frac{n_1 - n_2}{R}}$$

Refraction at a spherical surface

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

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Example 9.6 Light from a point source in air falls on a spherical glass surface ($n = 1.5$ and radius of curvature = 20 cm). The distance of the light source from the glass surface is 100 cm. At what position the image is formed?

Solution

We use the relation given by Eq. (9.16). Here

$u = -100$ cm, $v = ?$, $R = +20$ cm, $n_1 = 1$, and $n_2 = 1.5$.

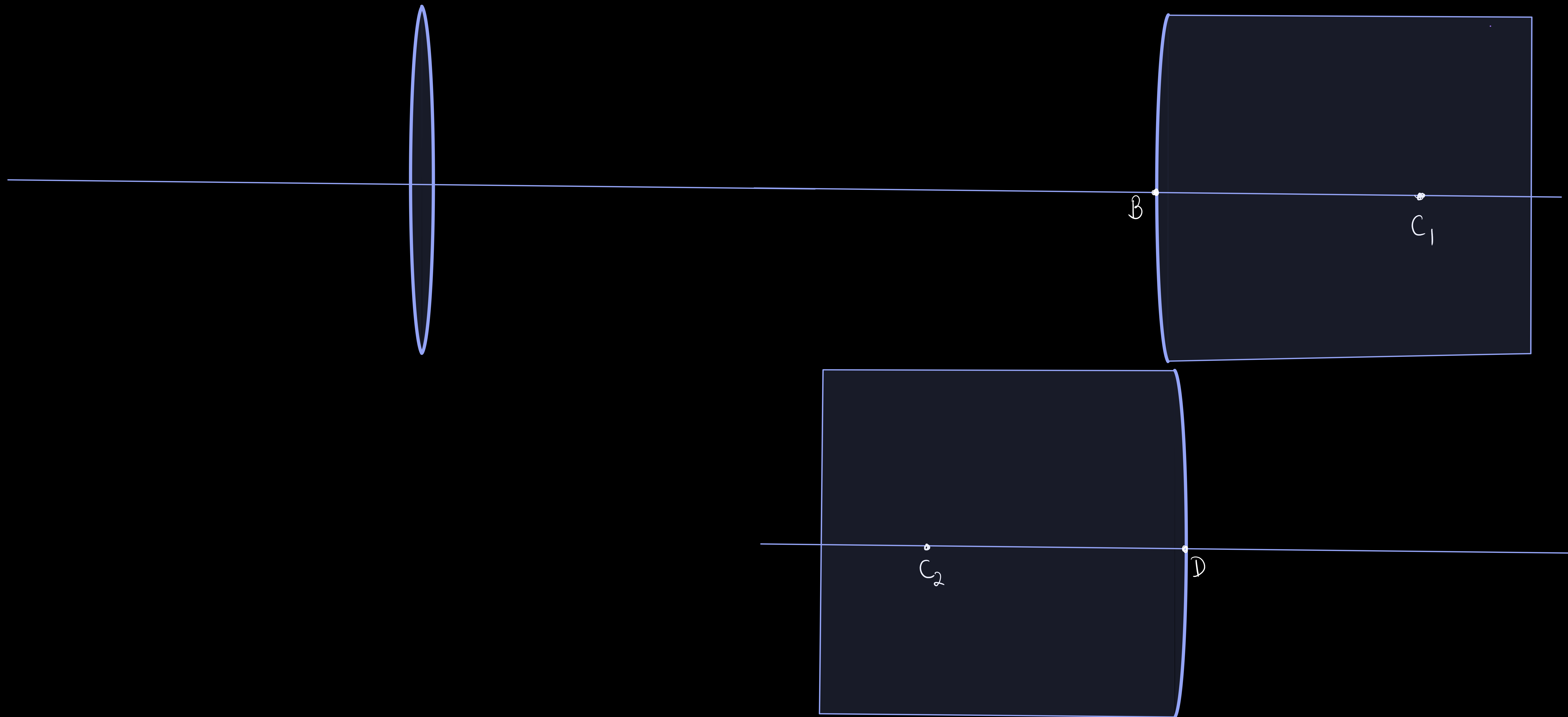
We then have

$$\frac{1.5}{v} + \frac{1}{100} = \frac{0.5}{20}$$

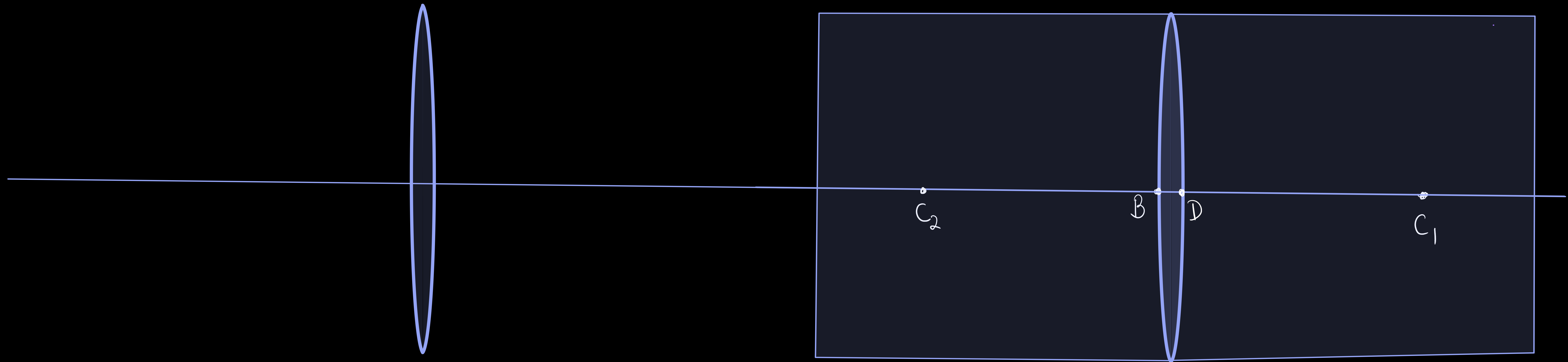
or $v = +100$ cm

The image is formed at a distance of 100 cm from the glass surface, in the direction of incident light.

Refraction by a lens (Lens Maker's Formula)

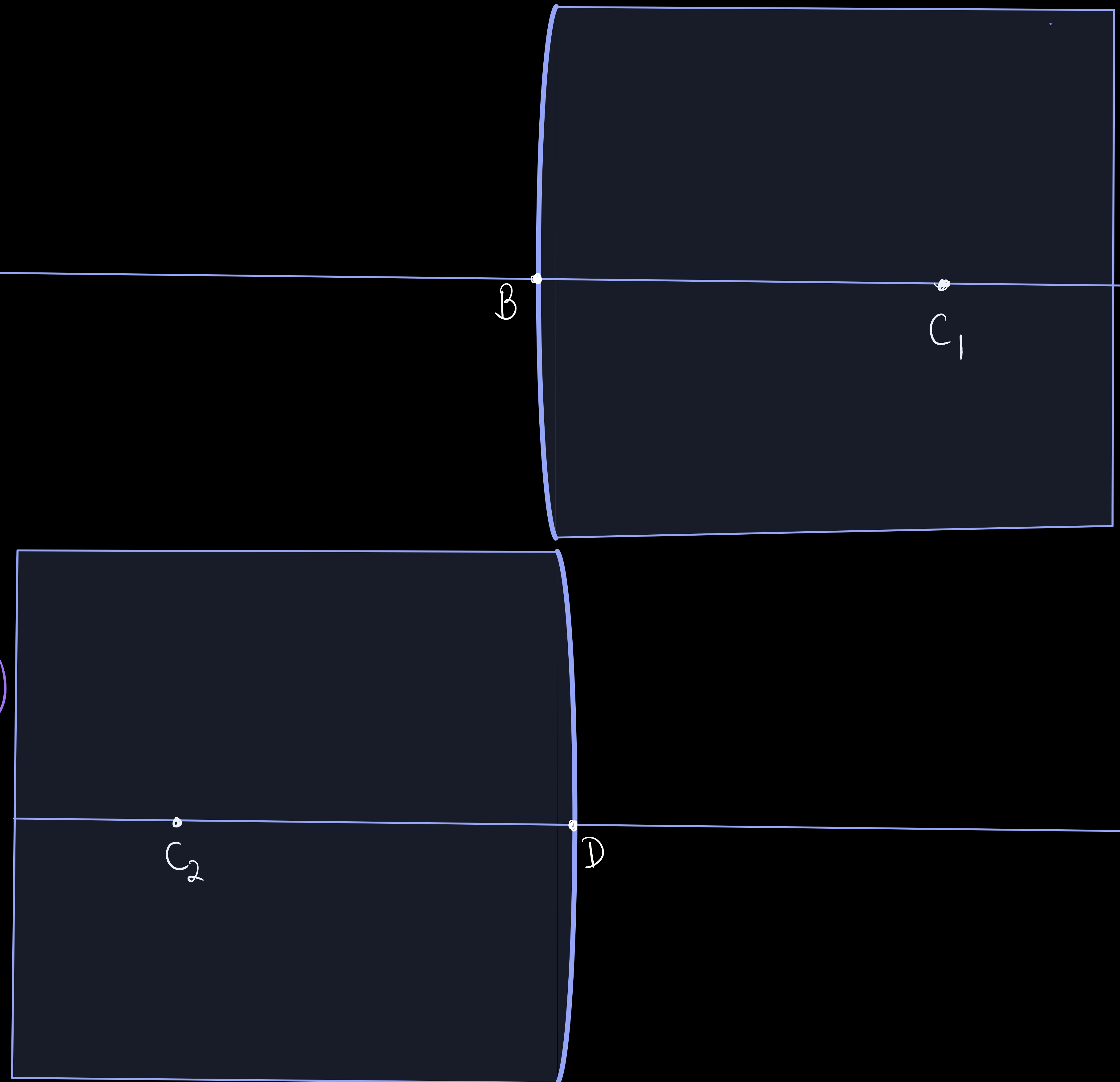
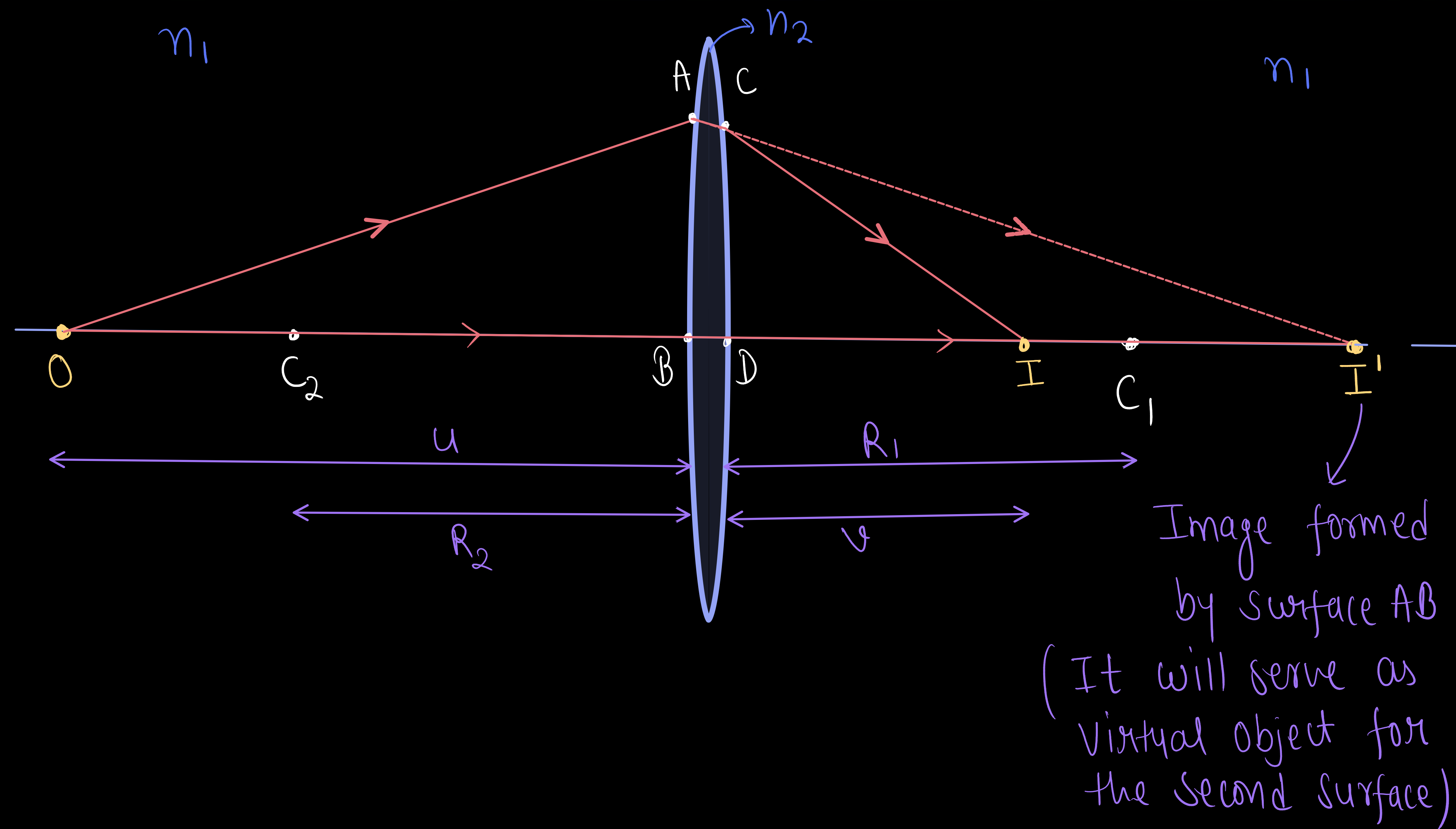


Refraction by a lens (Lens Maker's Formula)



- * A thin lens is a transparent optical medium bounded by two surfaces; at least one of which should be spherical.
- * Applying the formula for image formation by a single spherical surface successively at the two surfaces of a lens, we shall obtain the lens maker's formula and then the lens formula.

Refraction by a lens (Lens Maker's Formula)



Refraction by a lens (Lens Maker's Formula)

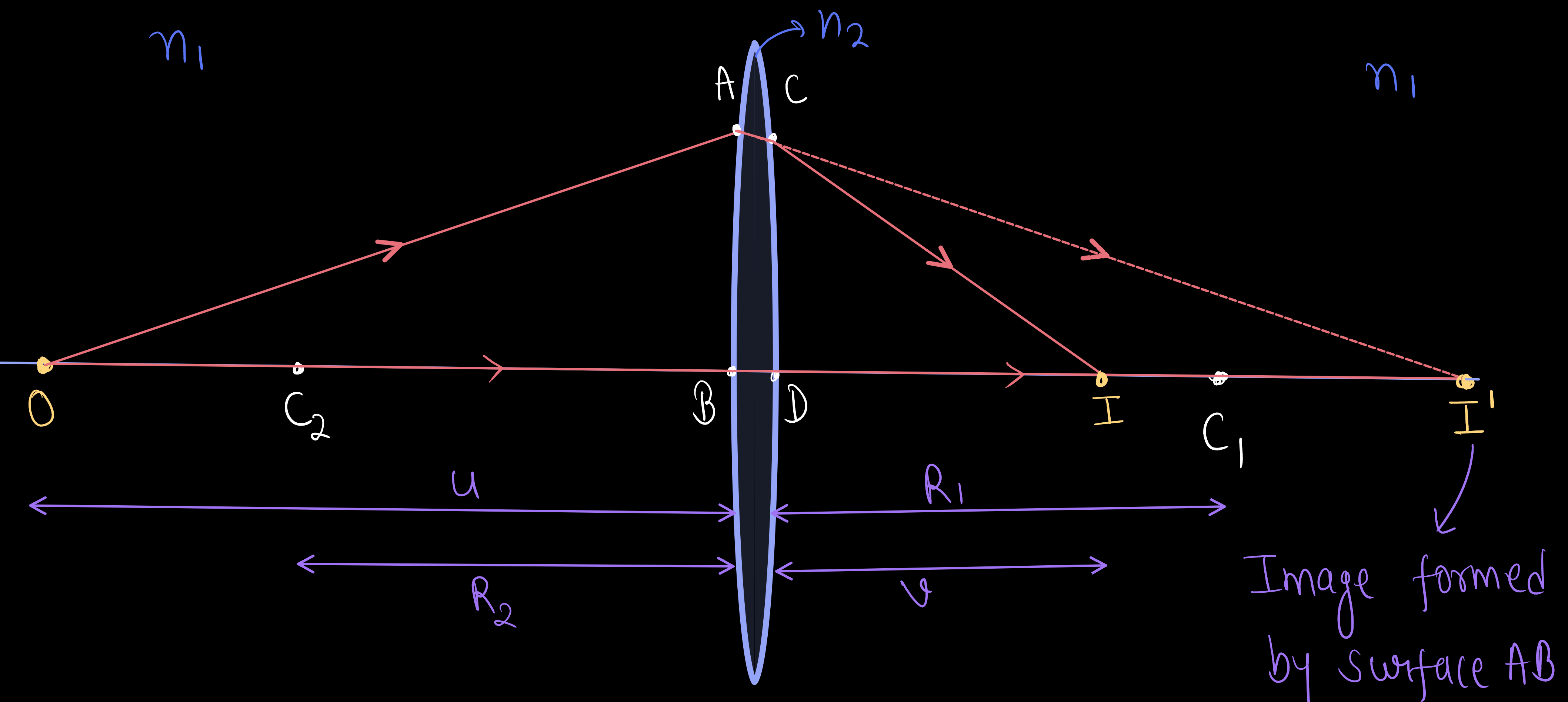
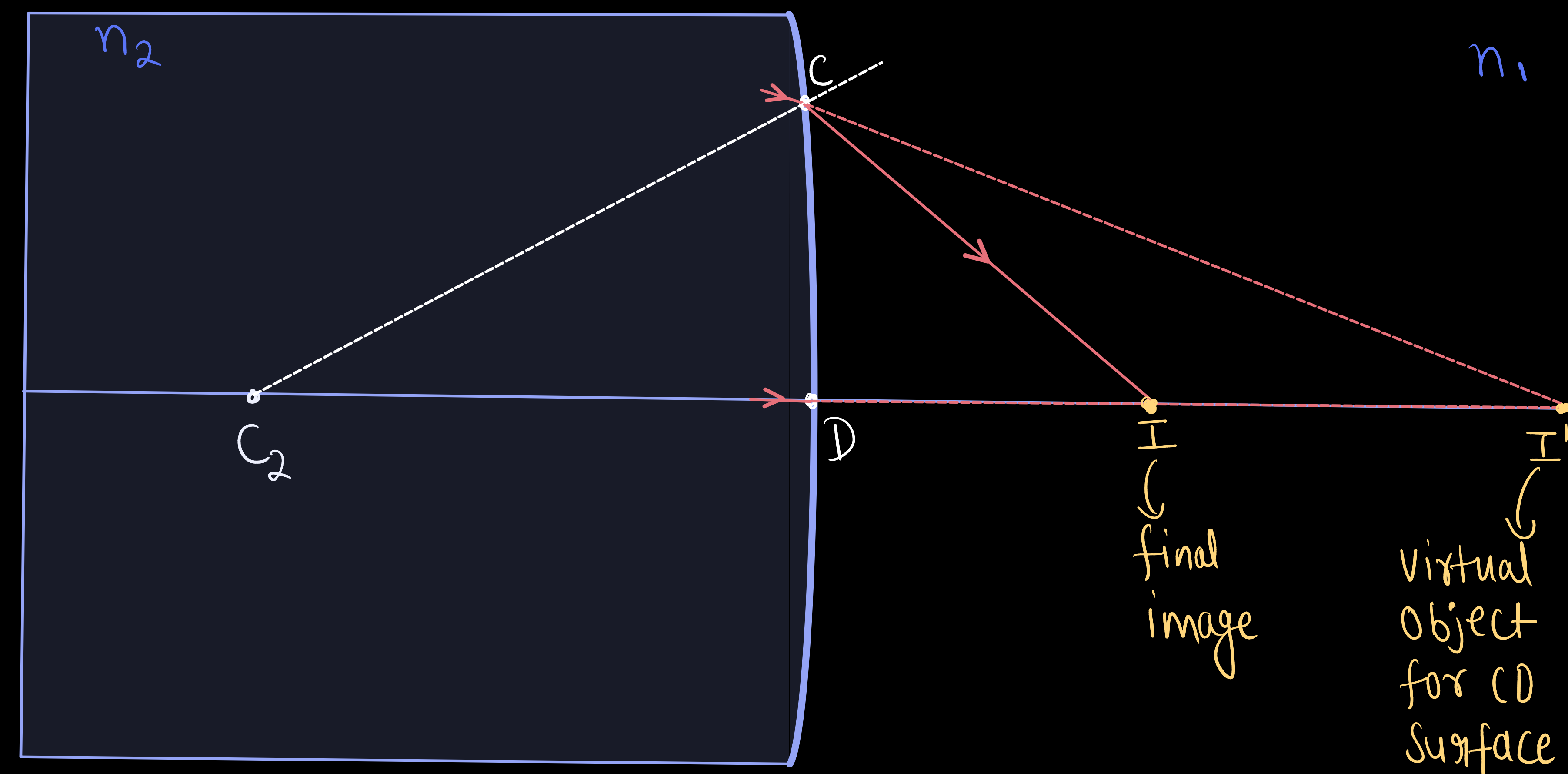
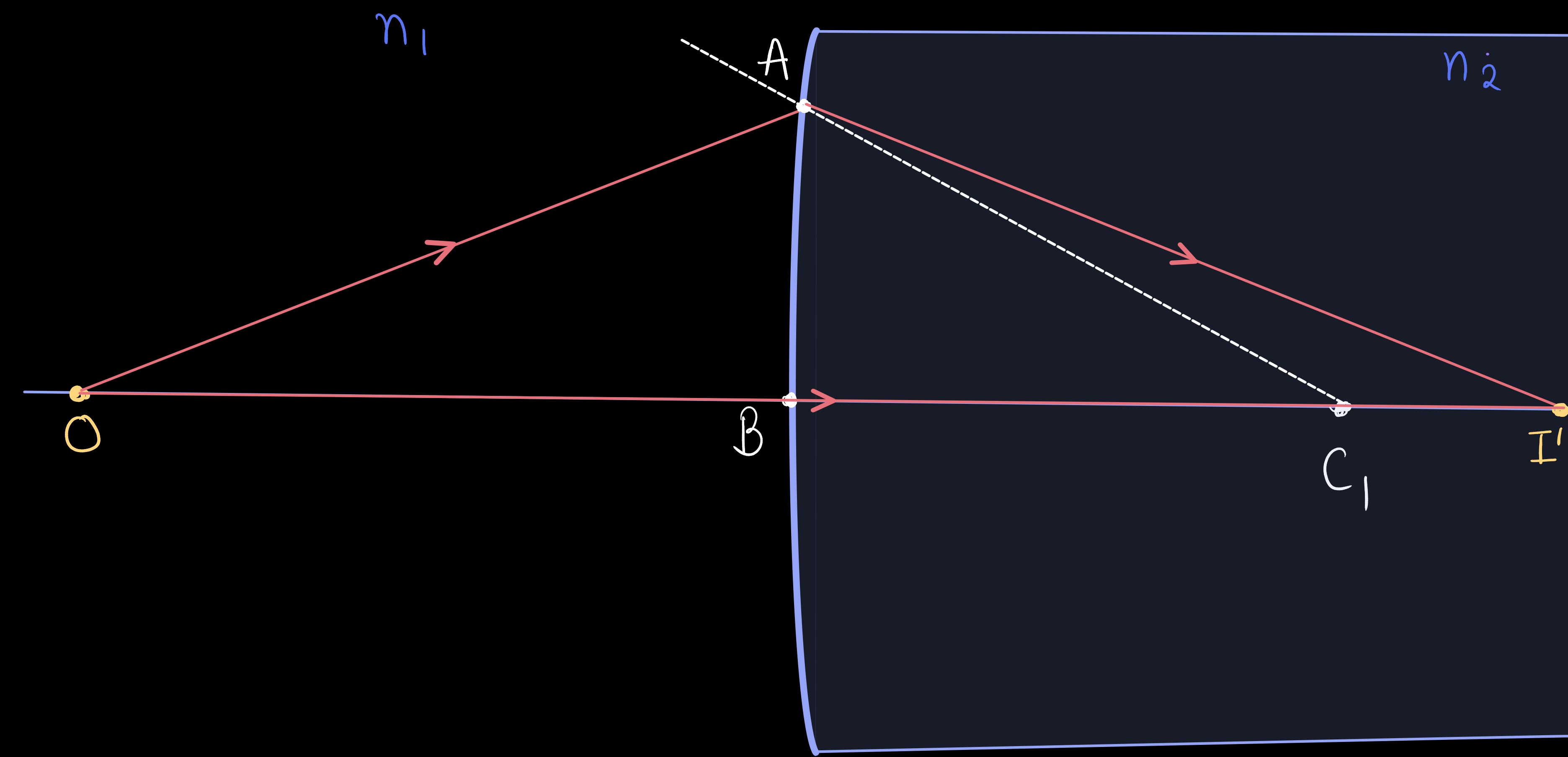


Image formed by surface AB
(It will serve as virtual object for the second surface)



* for first interface (AB) -

$$\frac{n_2}{BI'} - \frac{n_1}{(-BO)} = \frac{n_2 - n_1}{BC_1} \quad \text{--- (1)}$$

* for second interface (CD) -

$$\frac{n_1}{DI} - \frac{n_2}{DI'} = \frac{n_1 - n_2}{-DC_2} \quad \text{or} \quad \frac{n_1}{DI} - \frac{n_2}{BI'} = \frac{n_2 - n_1}{DC_2} \quad \text{--- (2)}$$

$\hookrightarrow (\because DI' \approx BI' \text{ for thin lens})$

Refraction by a lens (Lens Maker's Formula)

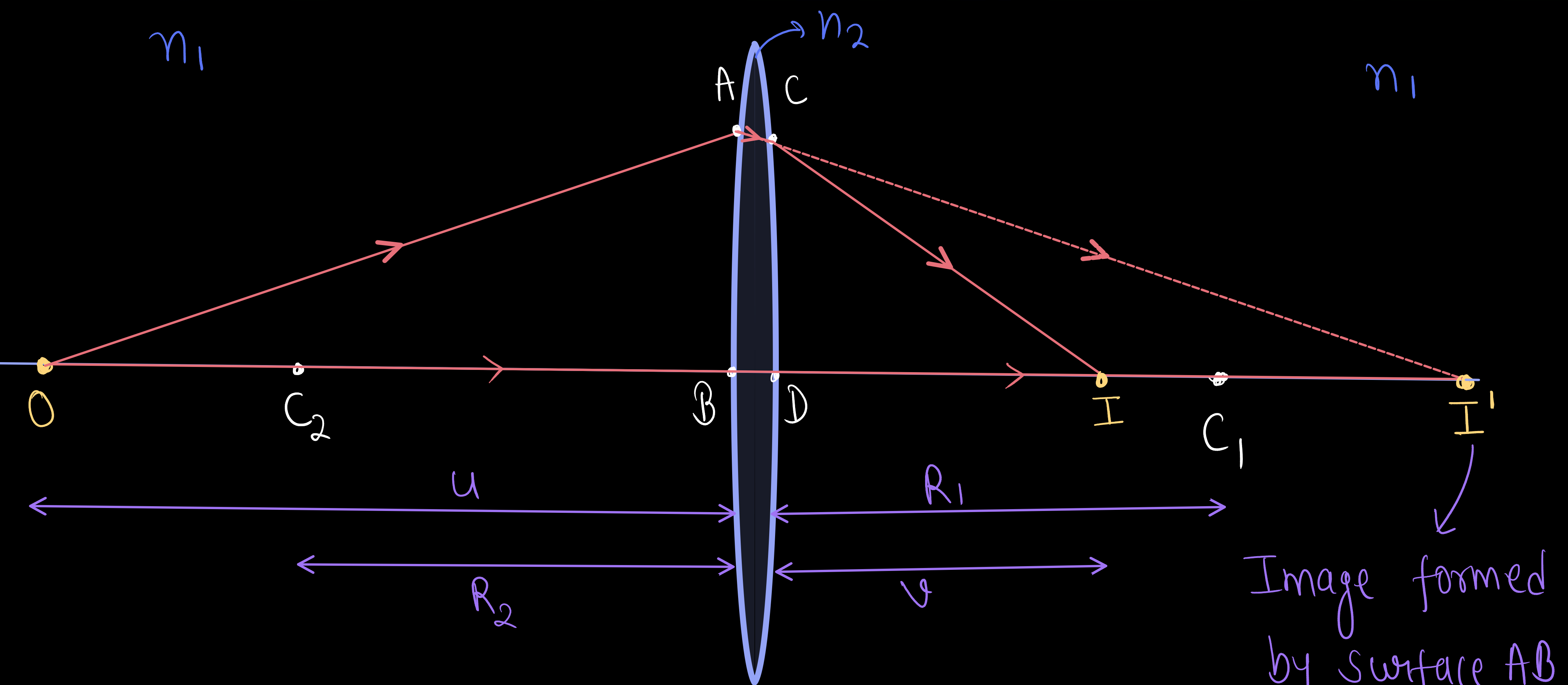


Image formed by surface AB
(It will serve as virtual object for the second surface)

* for first interface (AB) -

$$\frac{n_2}{BI'} - \frac{n_1}{(-BO)} = \frac{n_2 - n_1}{BC_1} \quad \text{--- (1)}$$

* for second interface (CD) -

$$\frac{n_1}{DI} - \frac{n_2}{DI'} = \frac{n_1 - n_2}{-DC_2} \quad \text{or} \quad \frac{n_1}{DI} - \frac{n_2}{BI'} = \frac{n_2 - n_1}{DC_2} \quad \text{--- (2)}$$

(∵ $DI' \approx BI'$ for thin lens)

Adding eq (1) & (2)

$$\frac{n_1}{DI} + \frac{n_1}{BO} = (n_2 - n_1) \left(\frac{1}{DC_1} + \frac{1}{BC_2} \right)$$

$$\frac{1}{DI} + \frac{1}{BO} = \left(\frac{n_2 - n_1}{n_1} \right) \left(\frac{1}{DC_1} + \frac{1}{BC_2} \right)$$

Now, using sign convention -

$$DI = v, \quad BO = -u, \quad DC_1 = R_1, \quad BC_2 = -R_2$$

$$\frac{1}{v} - \frac{1}{u} = \left(\frac{n_2}{n_1} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \text{--- (3)}$$

if $u = -\infty, v = f$

$$\frac{1}{f} - \frac{1}{\infty} = \left(\frac{n_2}{n_1} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\boxed{\frac{1}{f} = \left(\frac{n_2}{n_1} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)} \quad \text{--- (4)} \quad \text{Lens maker's formula.}$$

from eq (3) & (4) $\Rightarrow$ $\boxed{\frac{1}{v} - \frac{1}{u} = \frac{1}{f}} \quad \text{--- (5)} \quad \text{Lens formula.}$

Power of a lens

- * Power of a lens is a measure of the convergence or divergence, which a lens introduces in the light falling on it.
- * Clearly, a lens of shorter focal length bends the incident light more.

- * The power P of a lens is defined as the tangent of the angle by which it converges or diverges a beam of light parallel to the principal axis falling at unit distance from the optical centre

$$P = \tan \delta = \frac{1}{f}$$

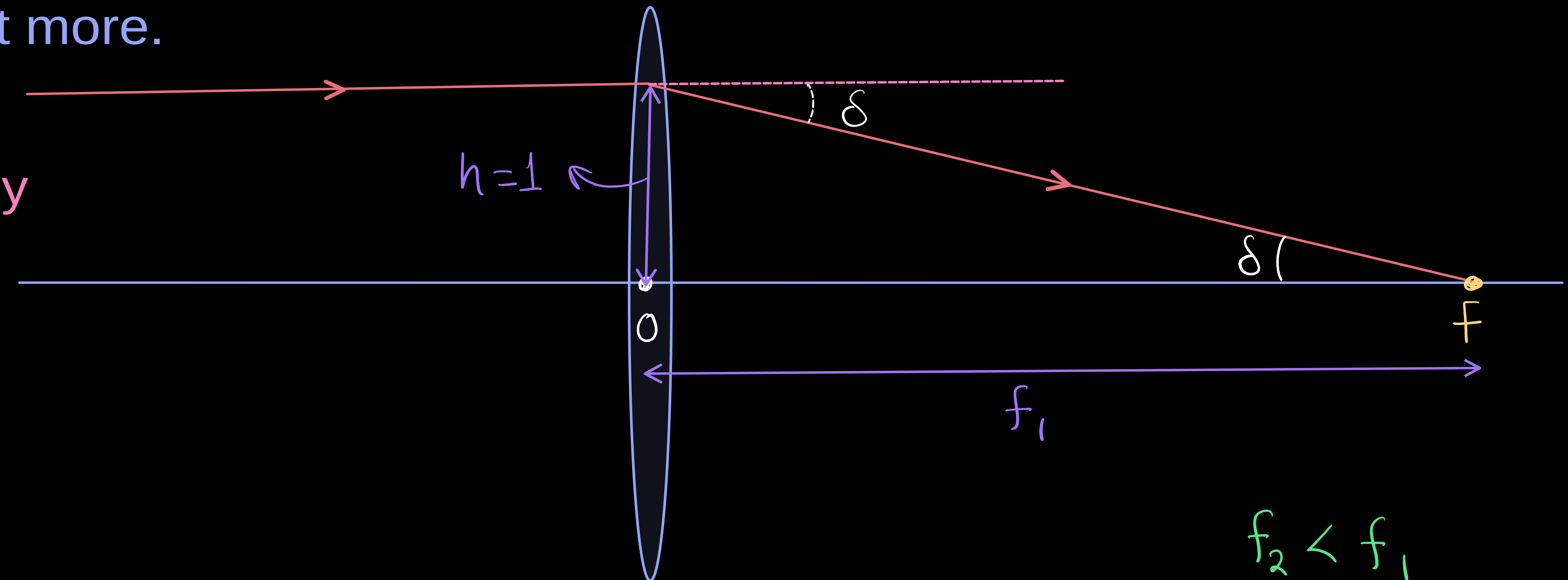
for small δ , $\tan \delta \approx \delta$

$$P = \delta = \frac{1}{f}$$

- * The SI unit for power of a lens is diopetre (D) : $1D = 1m^{-1}$

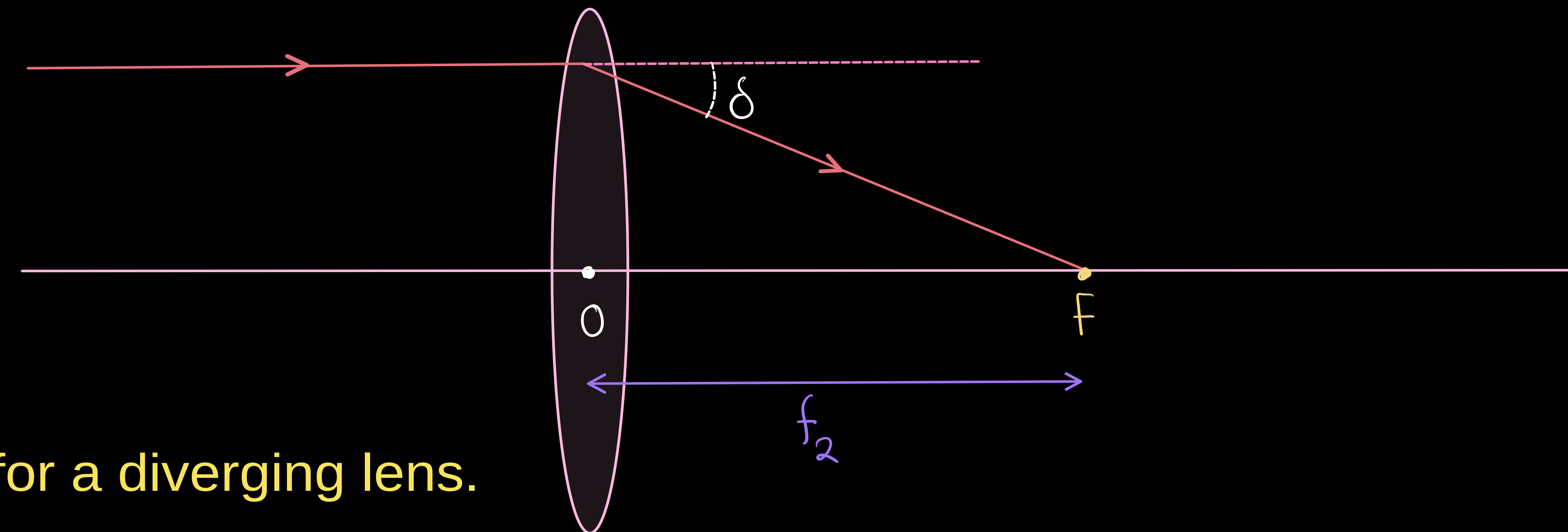
- * The power of a lens of focal length of 1 metre is one diopetre.

- * Power of a lens is positive for a converging lens and negative for a diverging lens.



$$f_2 < f_1$$

$$P_2 > P_1$$



Combination of thin lenses in contact

* Consider two lenses A and B of focal length f_1 and f_2 placed in contact with each other.

* The first lens A produces an image at I_1 .

* Since image I_1 is real, it serves as a virtual object for the second lens B, producing the final image at I.

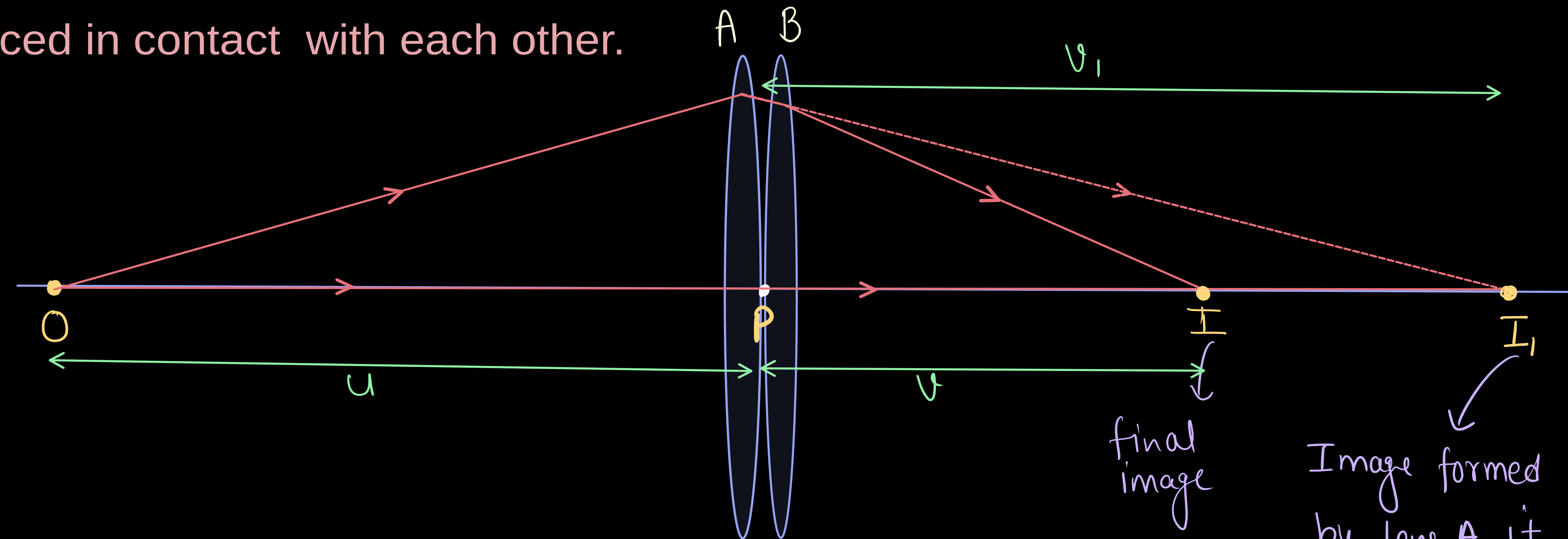


Image formed by first lens (A) -

$$\frac{1}{f_1} = \frac{1}{v_1} - \frac{1}{u} \quad \text{--- (1)}$$

Image formed by second lens (B) -

$$\frac{1}{f_2} = \frac{1}{v} - \frac{1}{v_1} \quad \text{--- (2)}$$

* Adding eq (1) and (2)

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{v} - \frac{1}{u} \quad \text{--- (3)}$$

* If the two lens-system is regarded as equivalent to a single lens of focal length f , we have

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \quad \text{--- (4)}$$

from eq (3) and (4) we get

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} \quad \text{--- (5)}$$

* Power of Combined lens -

$$P = P_1 + P_2 \quad \text{--- (6)}$$

* Total Magnification (m) of the combination -

$$m = m_1 \times m_2 \quad \text{--- (7)}$$

Image formed by lens A, it serves as virtual object for lens B

Example 9.9 Find the position of the image formed by the lens combination given in the Fig. 9.22.

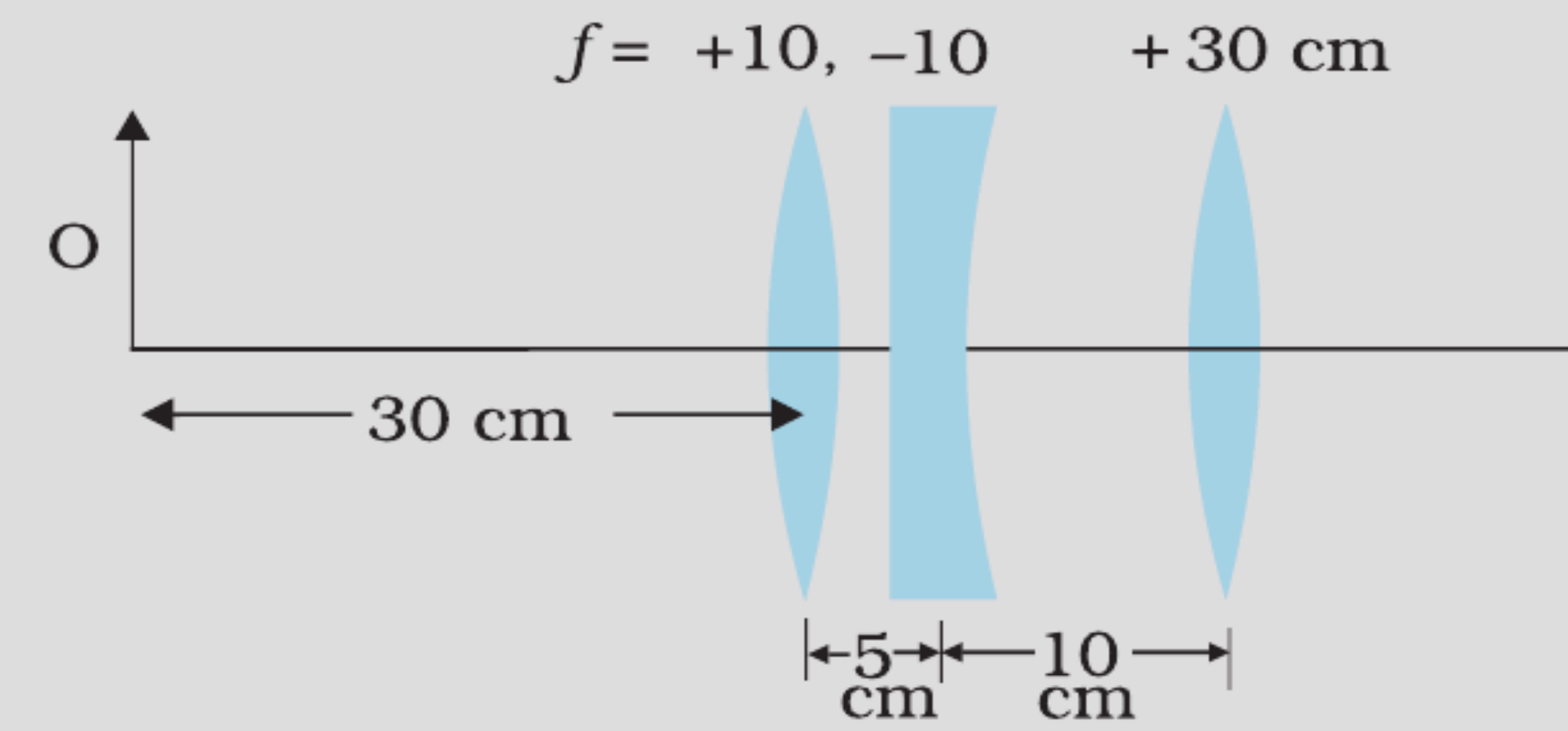


FIGURE 9.22

Solution Image formed by the first lens

$$\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1}$$

$$\frac{1}{v_1} - \frac{1}{-30} = \frac{1}{10}$$

$$\text{or } v_1 = 15 \text{ cm}$$

The image formed by the first lens serves as the object for the second. This is at a distance of $(15 - 5) \text{ cm} = 10 \text{ cm}$ to the right of the second lens. Though the image is real, it serves as a virtual object for the second lens, which means that the rays appear to come from it for the second lens.

$$\frac{1}{v_2} - \frac{1}{10} = \frac{1}{-10}$$

$$\text{or } v_2 = \infty$$

The virtual image is formed at an infinite distance to the left of the second lens. This acts as an object for the third lens.

$$\frac{1}{v_3} - \frac{1}{u_3} = \frac{1}{f_3}$$

$$\text{or } \frac{1}{v_3} = \frac{1}{\infty} + \frac{1}{30}$$

$$\text{or } v_3 = 30 \text{ cm}$$

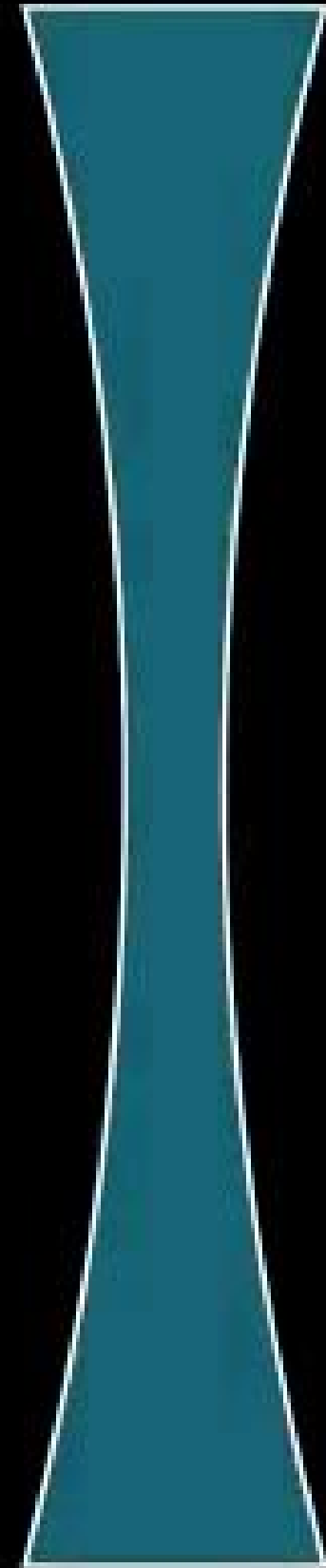
The final image is formed 30 cm to the right of the third lens.

Refraction by Spherical Lenses

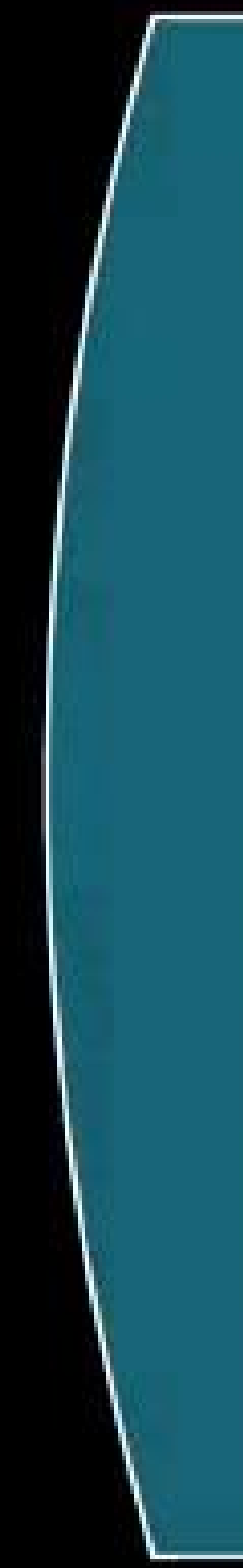
- **SPHERICAL LENS** : A spherical lens is a transparent material bounded by two surfaces one or both of which are spherical.
- A lens is bound by at least one spherical surface. In such lenses, the other surface would be plane. Spherical lenses are of four main types



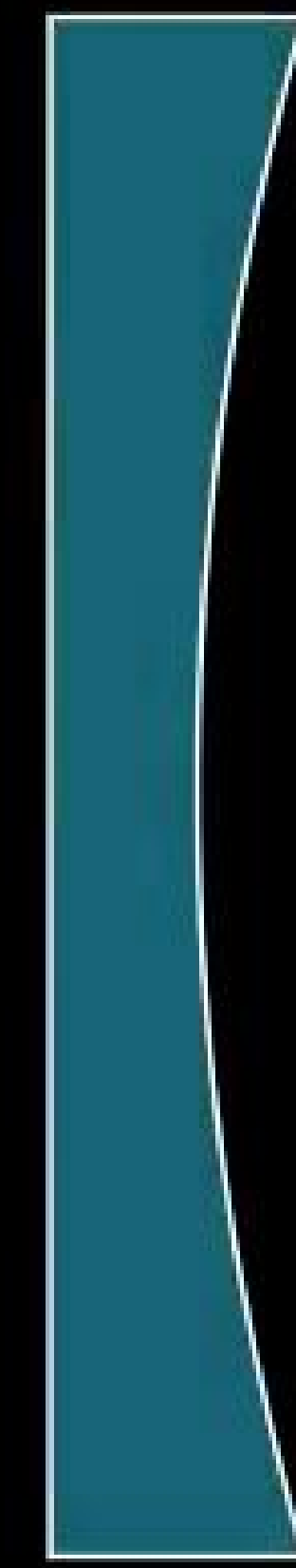
Double Convex Lens
(Or Convex Lens)



Double Concave Lens
(Or Concave Lens)



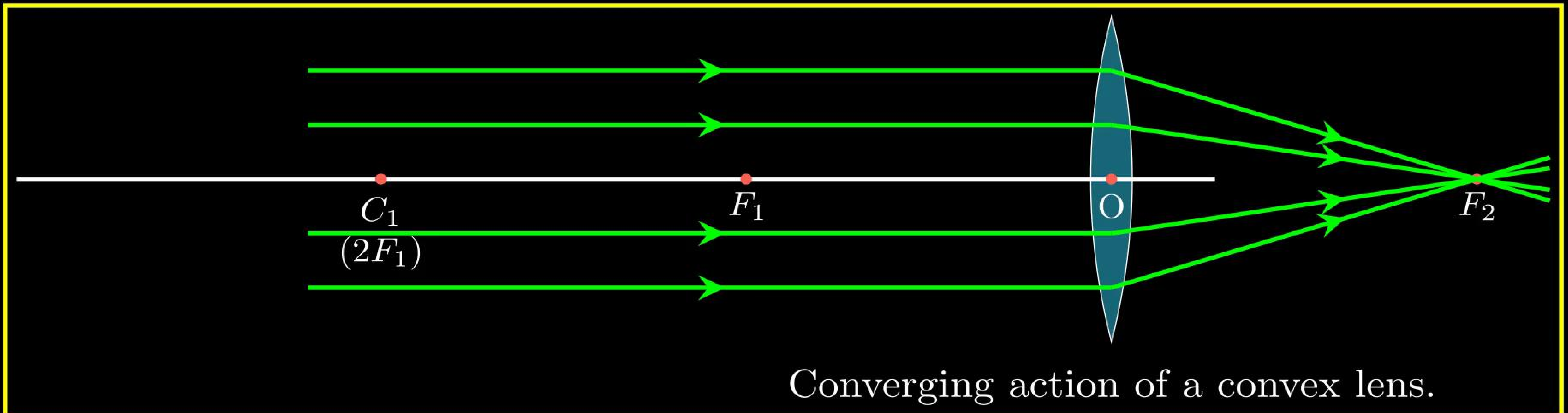
Plano-Convex Lens



Plano-Concave Lens

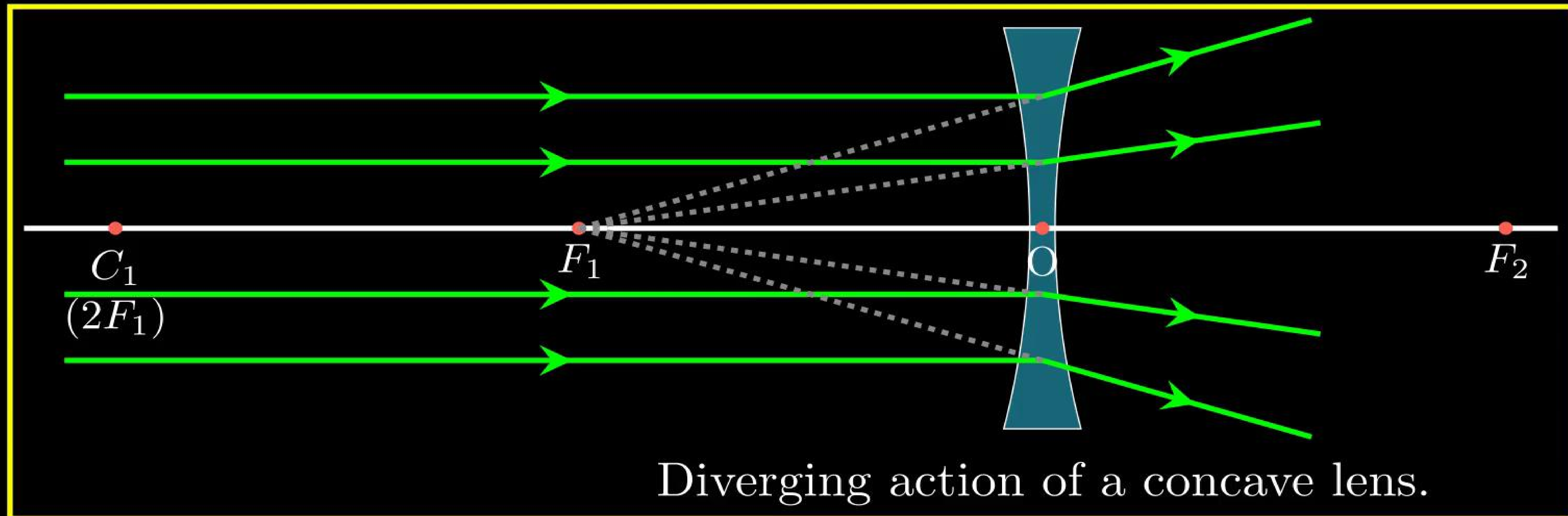
Refraction by Spherical Lenses

- **Convex lens** : A double convex lens is bounded by two spherical surfaces, **curved outwards (bulging outwards)**.
- It is **thicker in the middle** and **thinner at the edges**.
- Convex lens converges light rays. So, it is also called **converging lenses**.



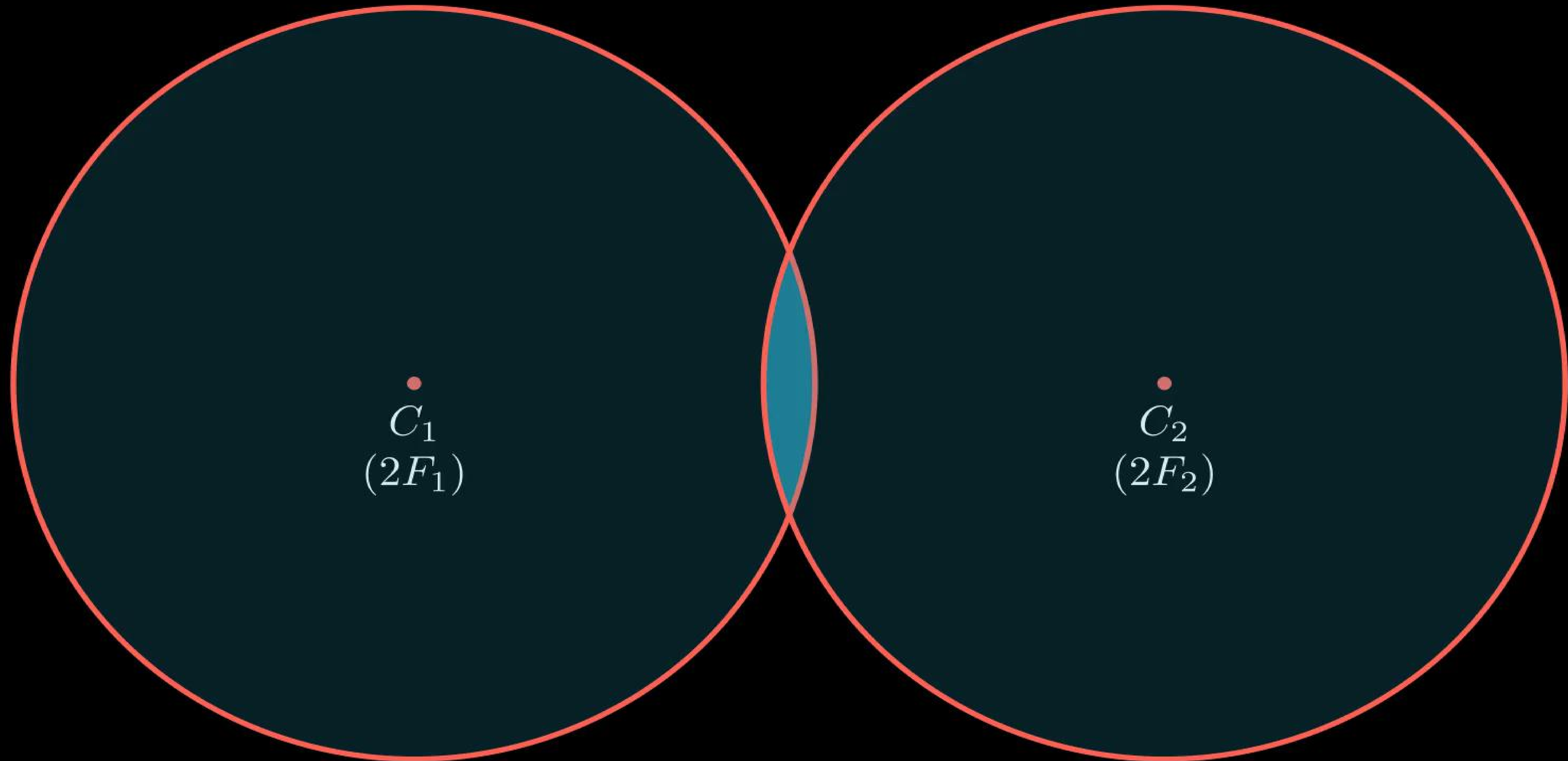
Refraction by Spherical Lenses

- **Concave lens** : A double convex lens is bounded by two spherical surfaces, **curved inwards**..
- It is **thinner in the middle** and **thicker at the edges**.
- Convex lens diverges light rays. So, it is also called **diverging lenses**.



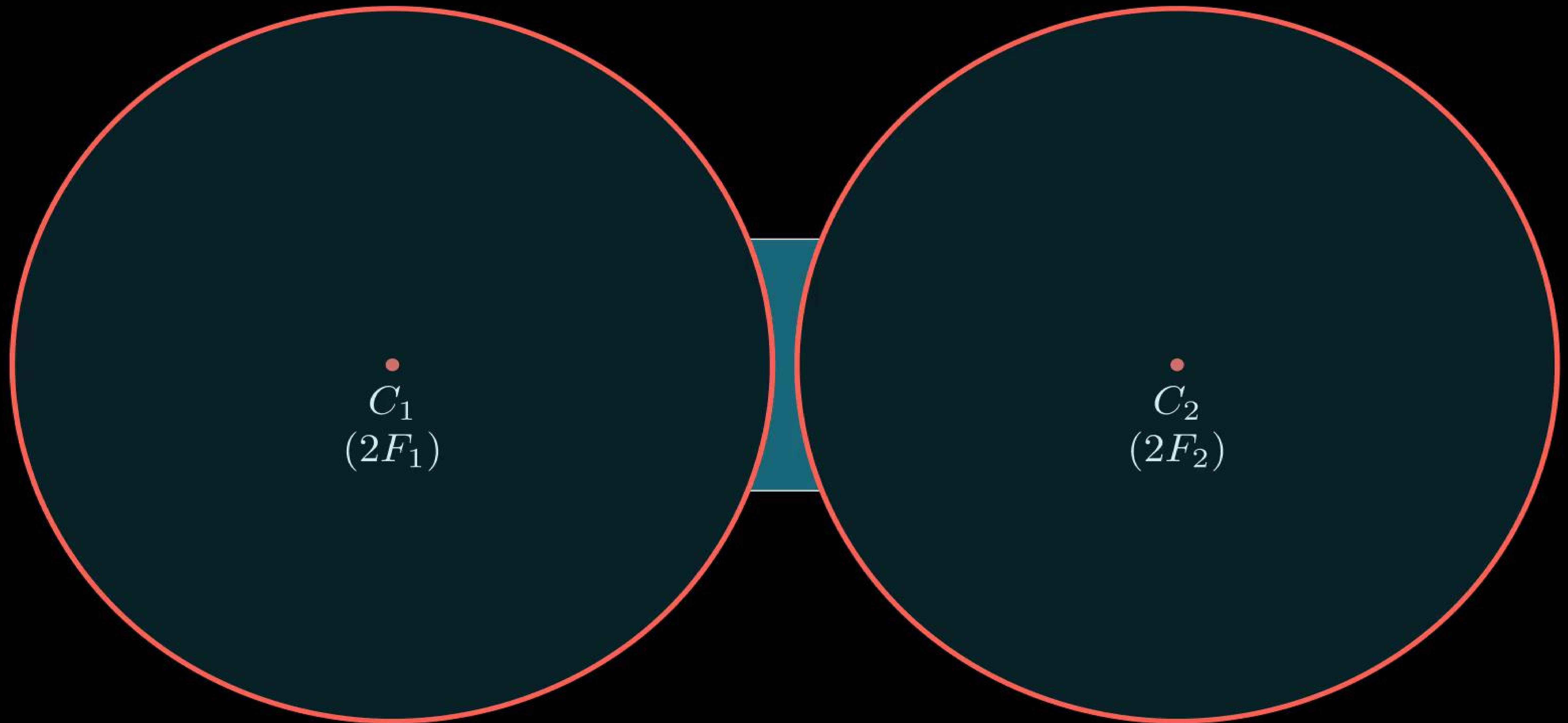
Important Terms Related to Spherical Lenses:

- (1) Centre of Curvature (C):



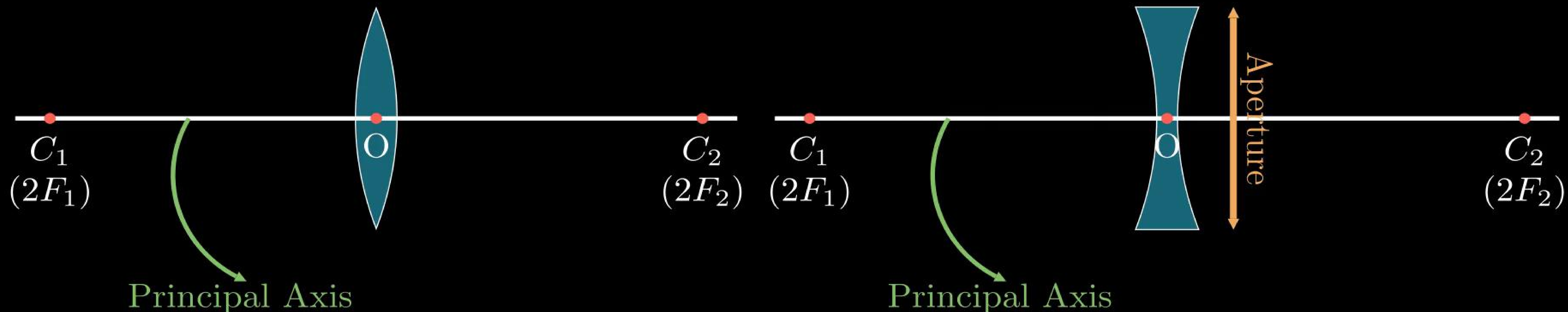
Important Terms Related to Spherical Lenses:

- (1) **Centre of Curvature (C):** The centres of the spheres that the spherical lens was a part of. A spherical lens has two centres of curvatures (C_1 , C_2).



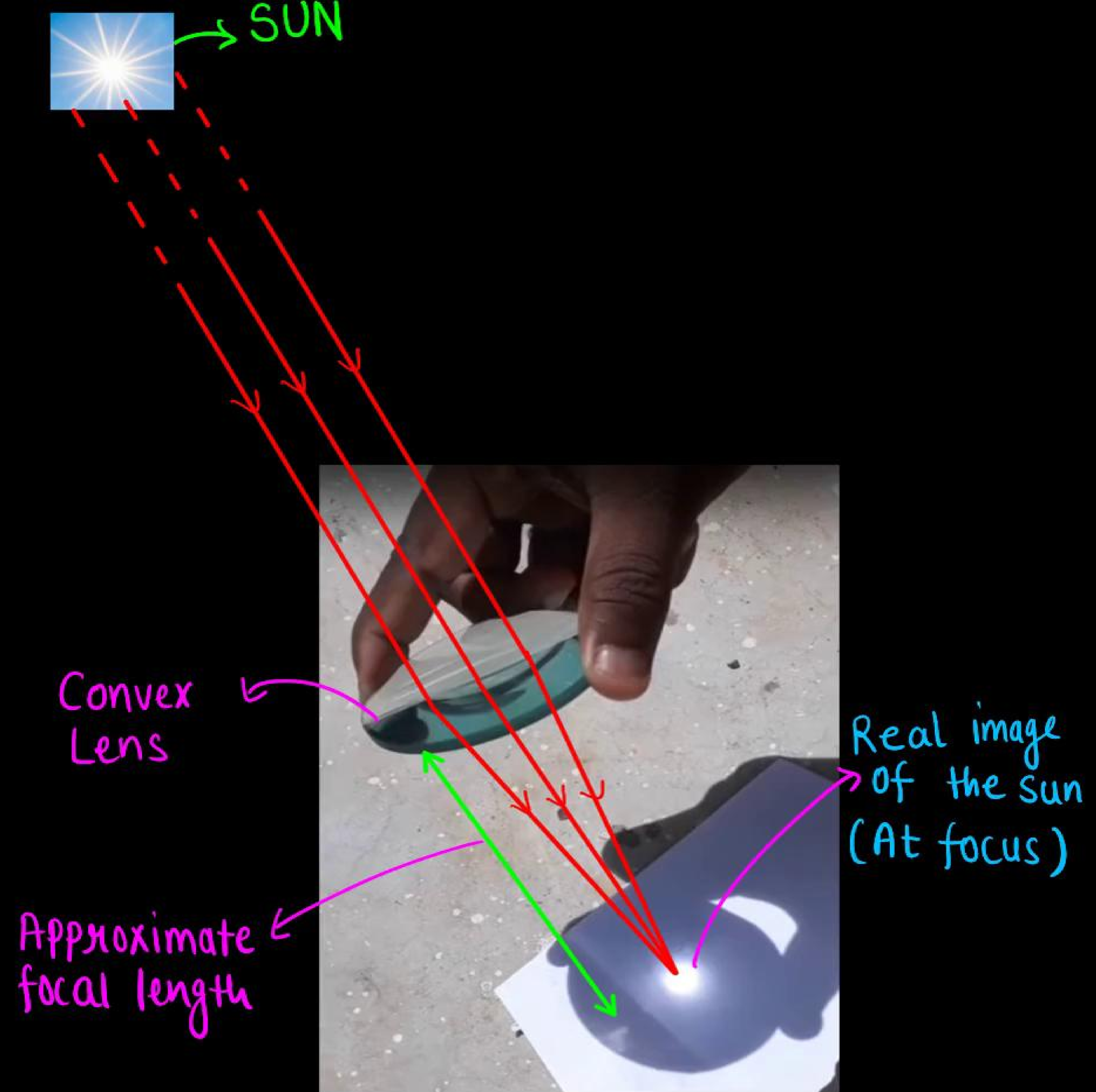
Important Terms Related to Spherical Lenses:

- **(2) Principal Axis:** An imaginary straight line passing through the two centres of curvature of a lens is called its principal axis.
- **(3) Optical Centre (O):** The central point of a lens is its optical centre. A ray of light through the optical centre of a lens passes without suffering any deviation.
- **(4) Aperture :** The effective diameter of the circular outline of a spherical lens is called its aperture. [Thin lenses $\rightarrow$ small aperture ($< R$) and $OC_1 = OC_2 = R$]

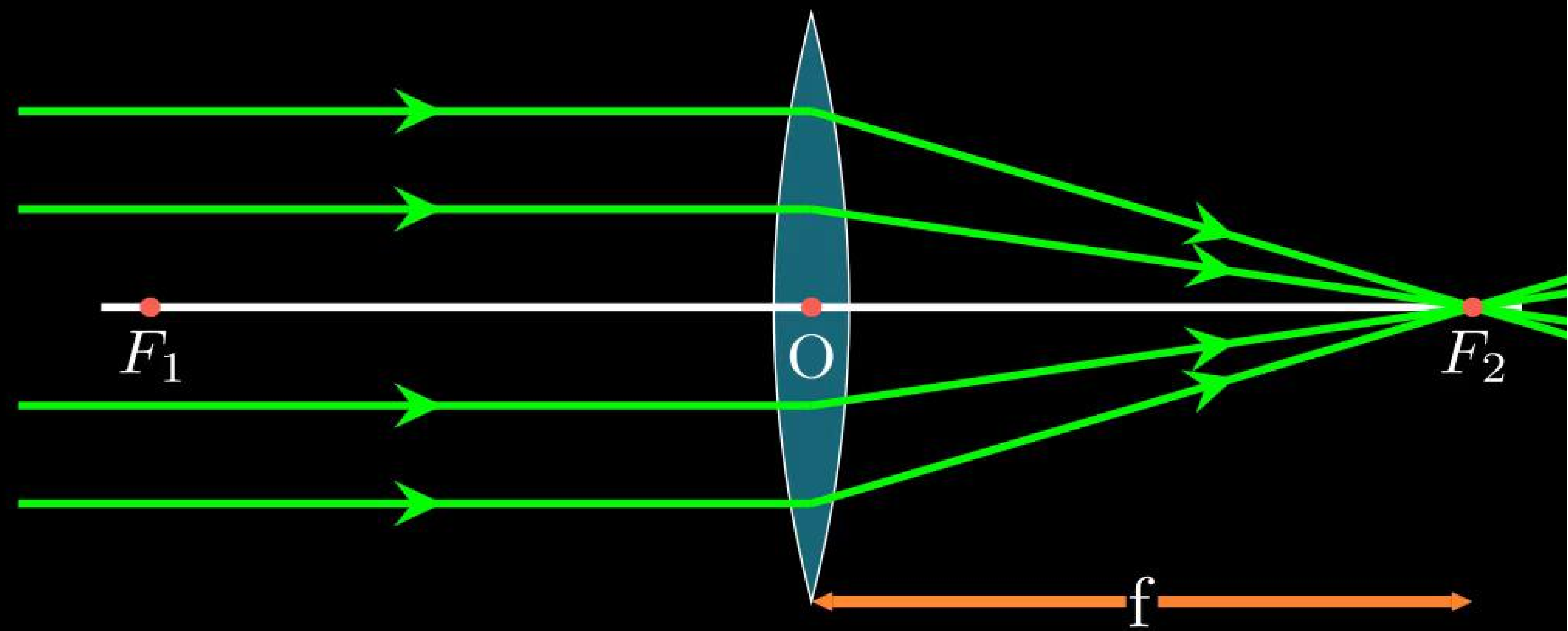


Activity 10.11:

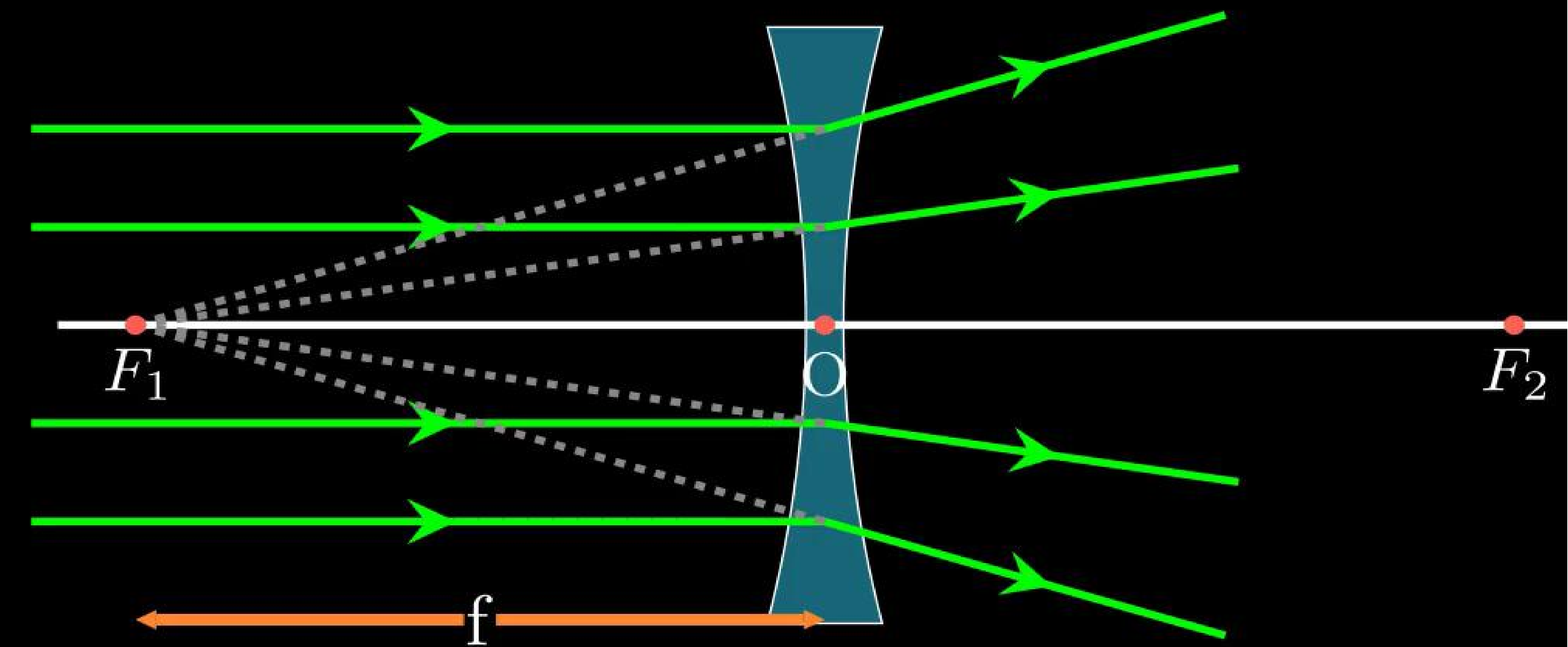
- Hold a convex lens in your hand. Direct it towards the Sun.
- Focus the light from the Sun on a sheet of paper. Obtain a sharp bright image of the Sun.
- Hold the paper and the lens in the same position for a while. Keep observing the paper.
- We observe that the paper begins to burn producing smoke.
- Parallel sun rays were converged by the lens at the sharp bright spot formed on the paper. The concentration of the sunlight at a point generated heat. This caused the paper to burn.



- **(5) Principal Focus of Convex Lens :**
When several rays of light parallel to the principal axis are falling on a convex lens. These rays, after refraction from the lens, are converging to a point on the principal axis. This point on the principal axis is called the principal focus of the lens



- **(6) Principal Focus of Concave Lens :**
When several rays of light parallel to the principal axis are falling on a concave lens. These rays, after refraction from the lens, are appearing to diverge from a point on the principal axis. This point on the principal axis is called the principal focus of the lens



- **Focal Length (f) :** distance of the principal focus from the optical centre.

Image Formation by Lenses and Ray Diagrams

Activity 10.12 : Image formation by a convex lens for different positions of the object

	Position of the Object	Position of the Image	Size of the Image	Nature of the Image
(1)	At infinity	At focus F_2	Highly diminished, point-sized	Real and inverted
(2)	Beyond C_1 ($2F_1$)	Between F_2 and C_2 ($2F_2$)	Diminished	Real and inverted
(3)	At C_1 ($2F_1$)	At C_2 ($2F_2$)	Same size	Real and inverted
(4)	Between C_1 ($2F_1$) and F_1	Beyond C_2 ($2F_2$)	Enlarged	Real and inverted
(5)	At F_1	At infinity	Highly enlarged	Real and inverted
(6)	Between O and F_1	On the same side of the lens as the object	Enlarged	Virtual and erect

Image Formation by Lenses and Ray Diagrams

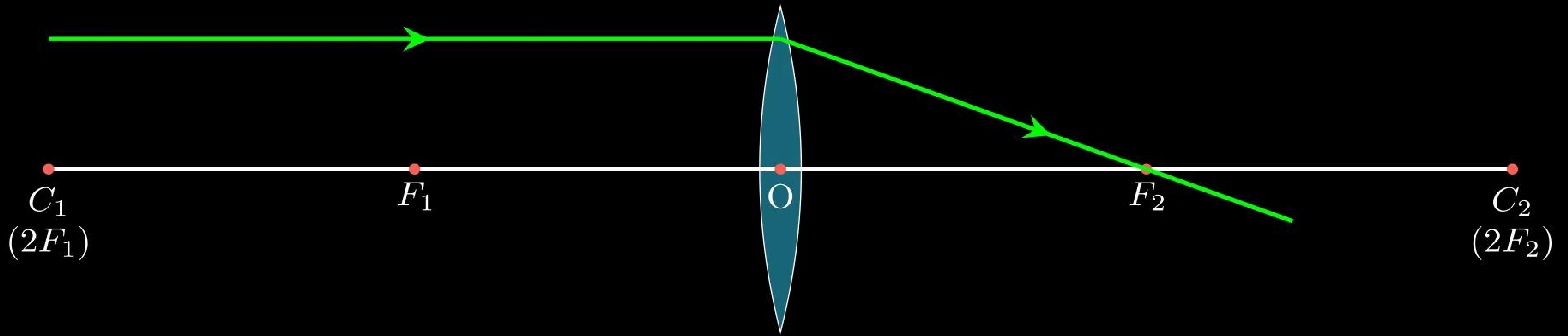
Activity 10.13 : Image formation by a concave lens for different positions of the object

	Position of the Object	Position of the Image	Size of the Image	Nature of the Image
(1)	At infinity	At focus F_1	Highly diminished, point-sized	Virtual and erect
(2)	Between infinity and optical centre O of the lens	Between focus F_1 and optical centre O	Diminished	Virtual and erect

Note: A concave lens will always give a virtual, erect and diminished image, irrespective of the position of the object.

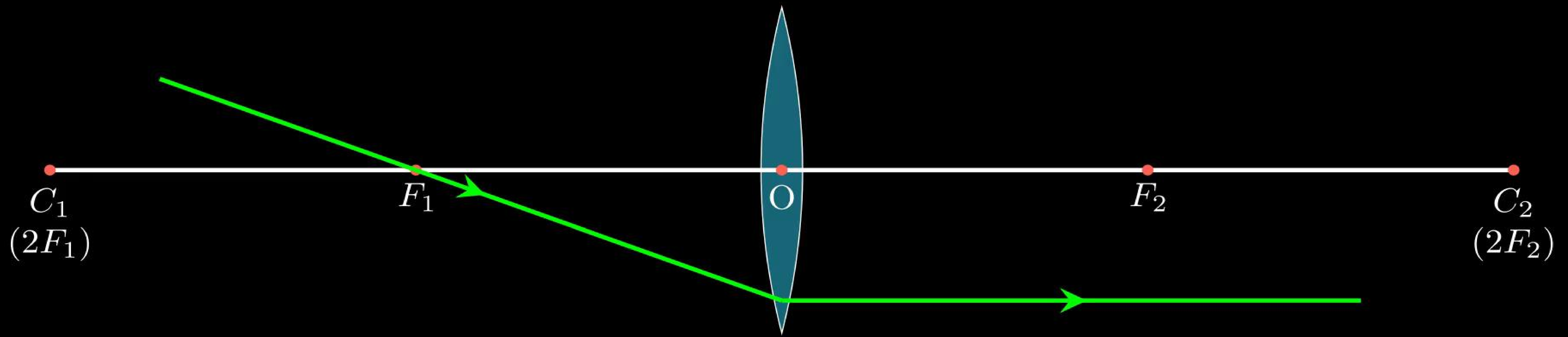
Ray Diagram for Convex Lens

(1) A ray of light parallel to the principal axis of a convex lens, passes through the principal focus on the other side of the lens.



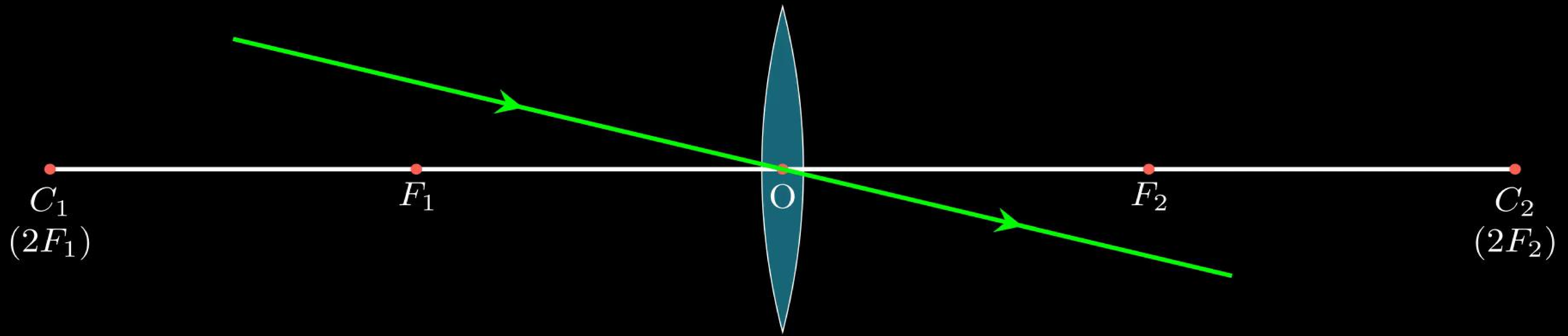
Ray Diagram for Convex Lens

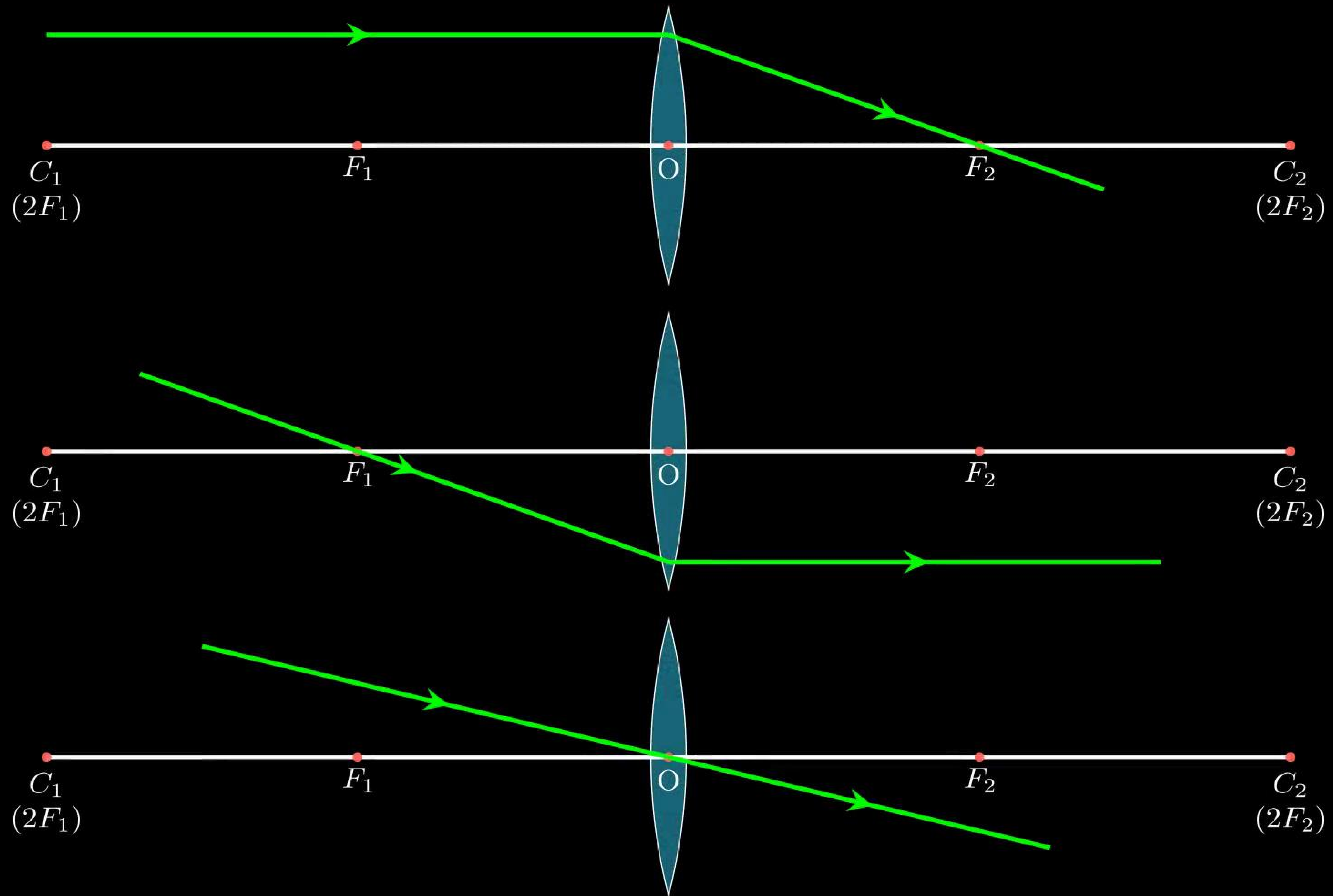
(2) A ray of light passing through a principal focus, after refraction from a convex lens, will emerge parallel to the principal axis.



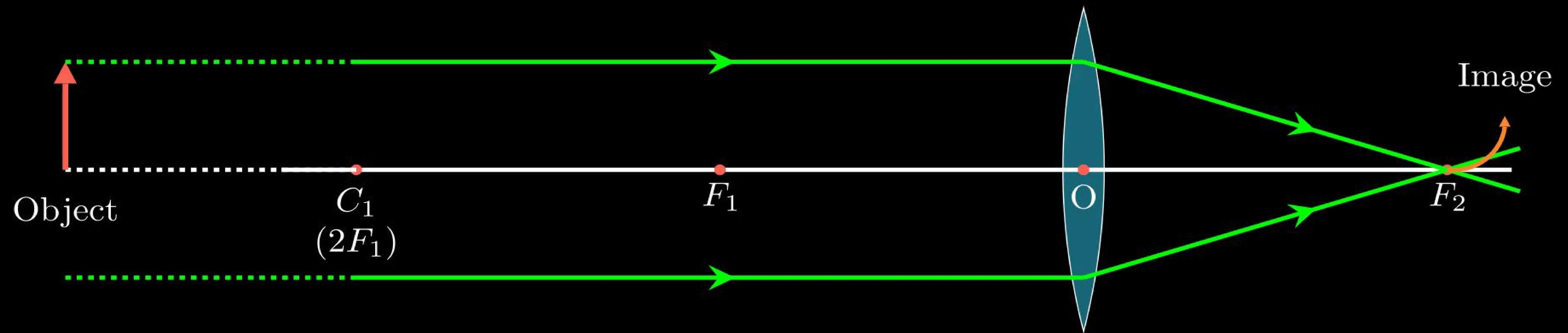
Ray Diagram for Convex Lens

(3) A ray of light passing through the optical centre of a lens will emerge without any deviation.



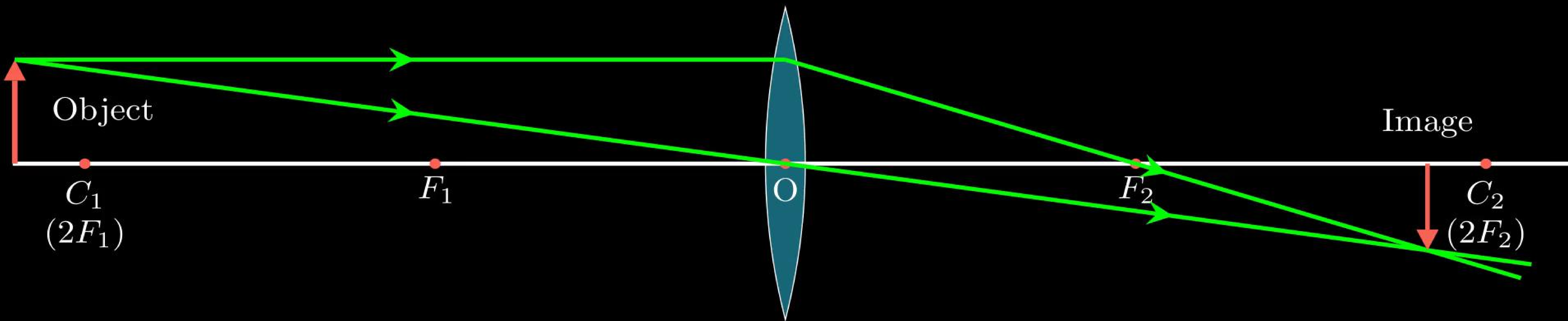


Position of Object (i): At infinity



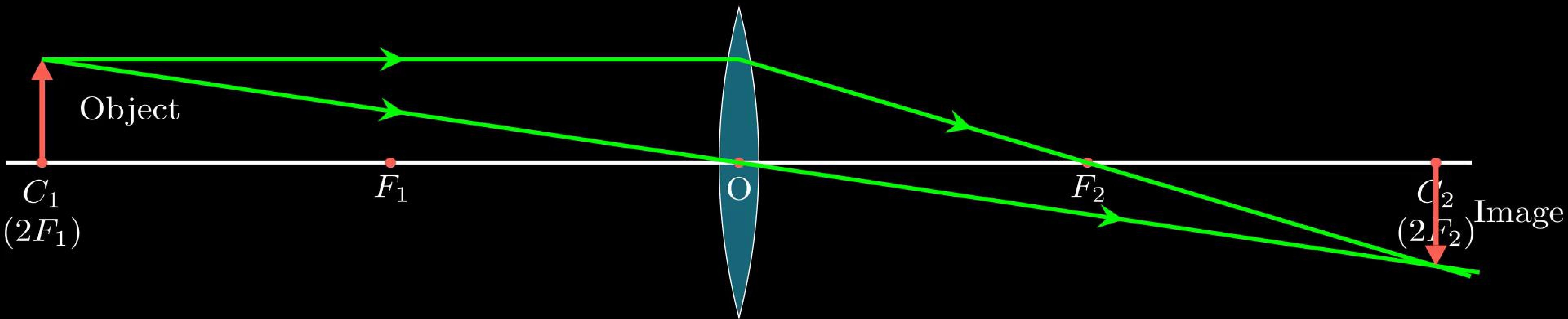
Position of the Image	Size of the Image	Nature of the Image
At the focus (F_2)	Highly diminished, point-sized	Real and inverted

Position of Object (ii): Beyond C_1 ($2F_1$)



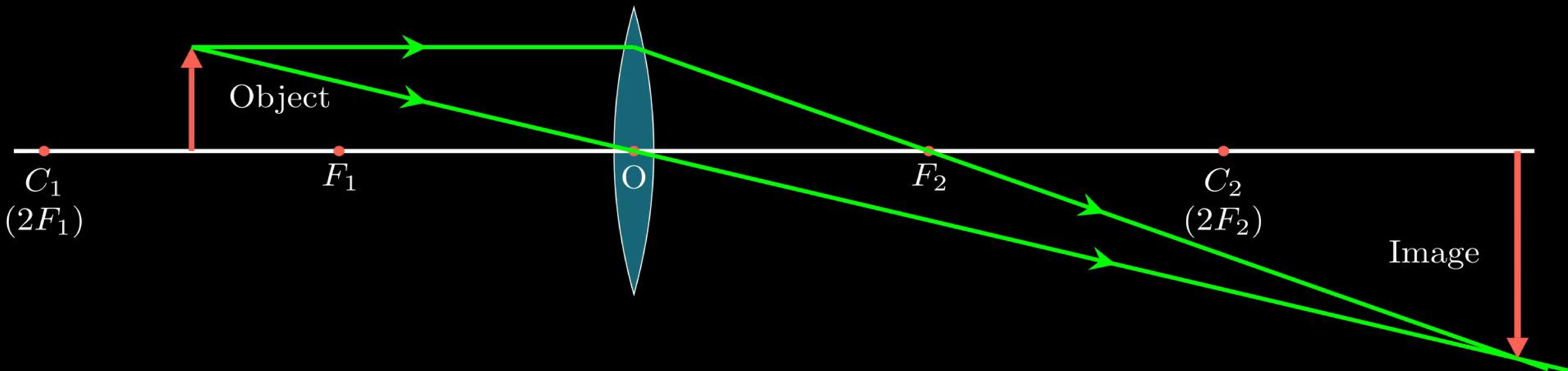
Position of the Image	Size of the Image	Nature of the Image
Between F_2 and $C_2(2F_2)$	Diminished	Real and inverted

Position of Object (iii): At C_1 ($2F_1$)



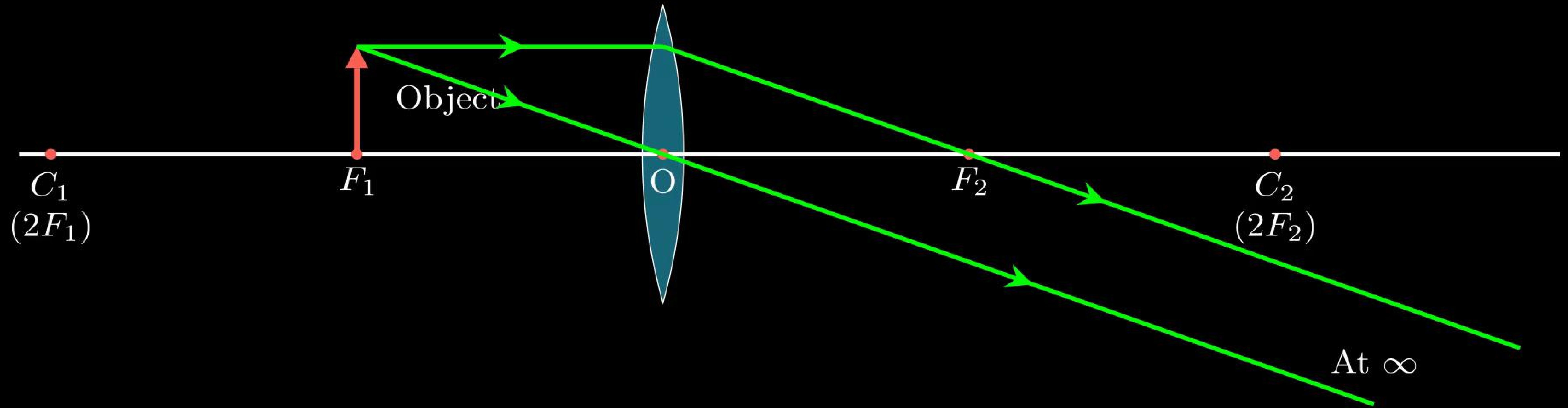
Position of the Image	Size of the Image	Nature of the Image
At $C_2(2F_2)$	Same Size	Real and inverted

Position of Object (iv): Between C_1 and F_1



Position of the Image	Size of the Image	Nature of the Image
Beyond C_2 ($2F_2$)	Enlarged	Real and inverted

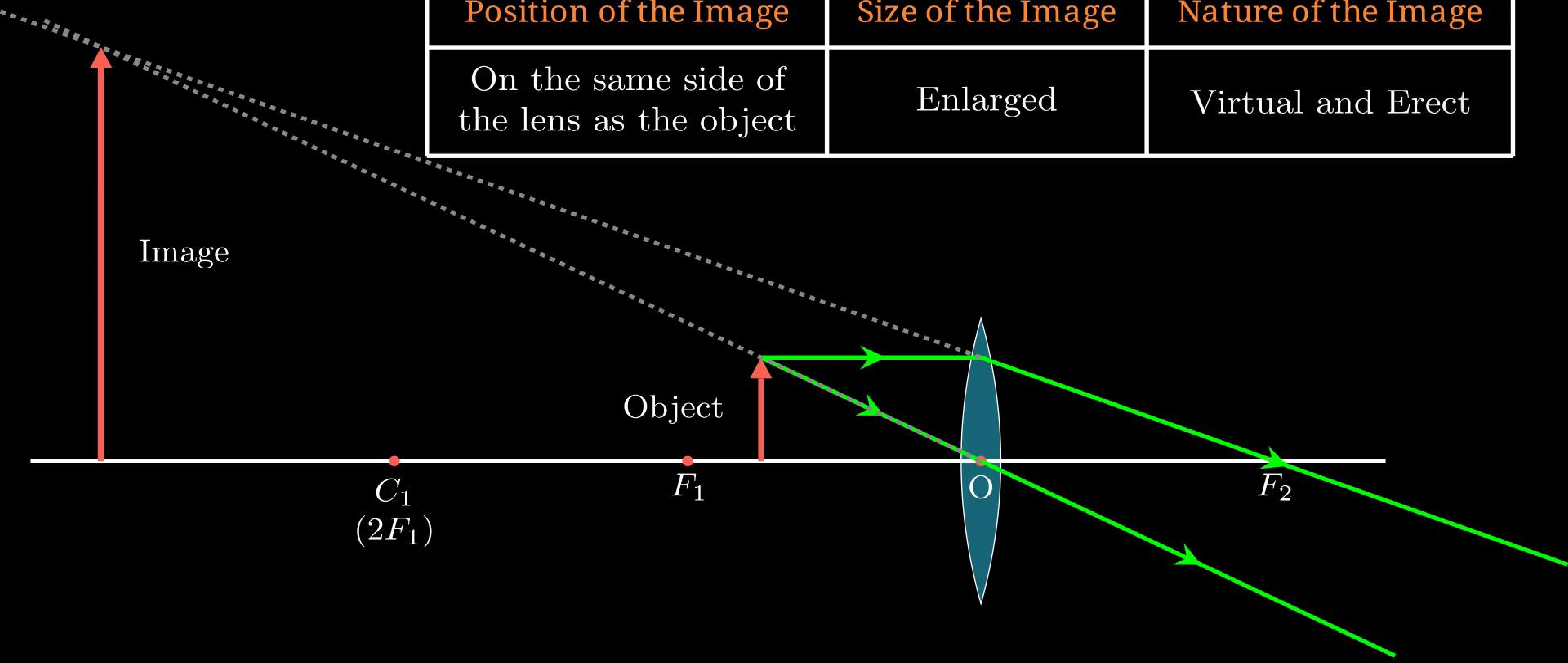
Position of Object (v): At F_1



Position of the Image	Size of the Image	Nature of the Image
At infinity (∞)	Highly Magnified (Highly Enlarged)	Real and inverted

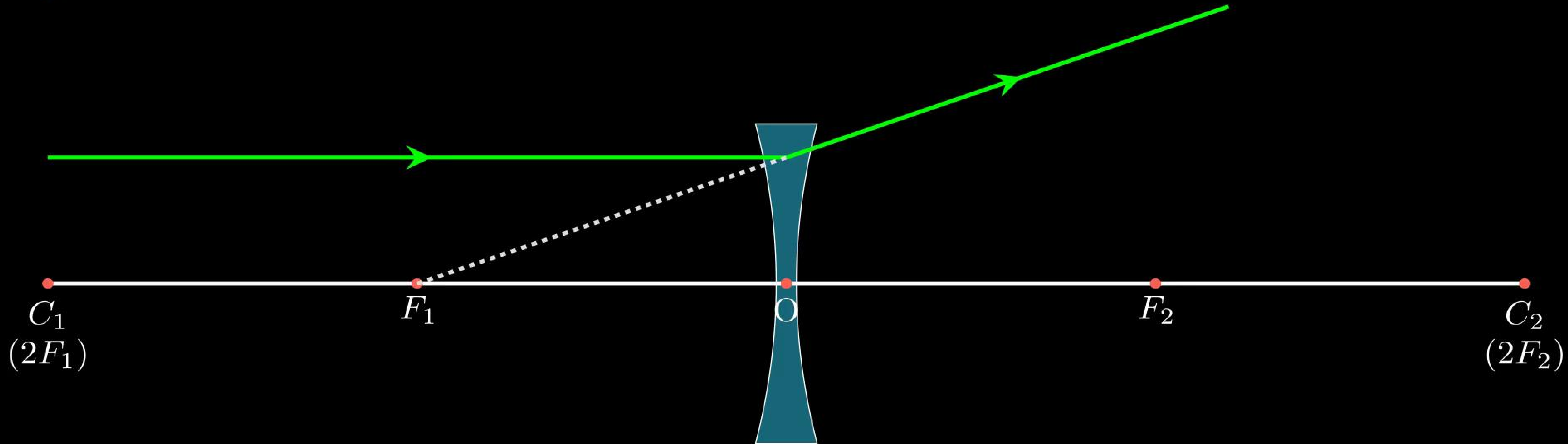
Position of Object (vi): Between F_1 and O

Position of the Image	Size of the Image	Nature of the Image
On the same side of the lens as the object	Enlarged	Virtual and Erect



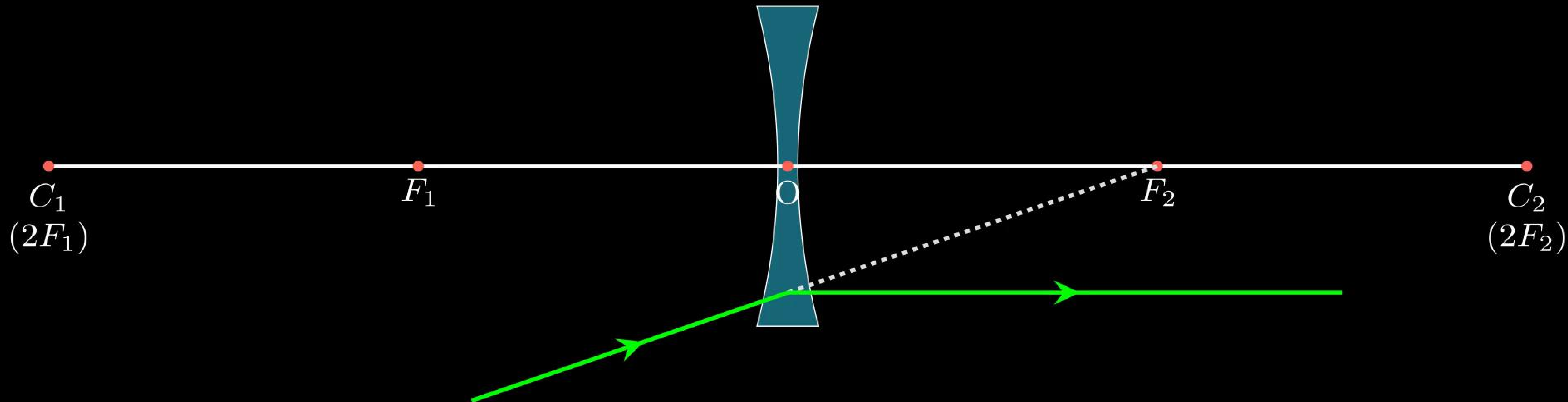
Ray Diagram for Concave Lens

(1) A ray of light parallel to the principal axis of a concave lens, after refraction the ray appears to diverge from the principal focus located on the same side of the lens



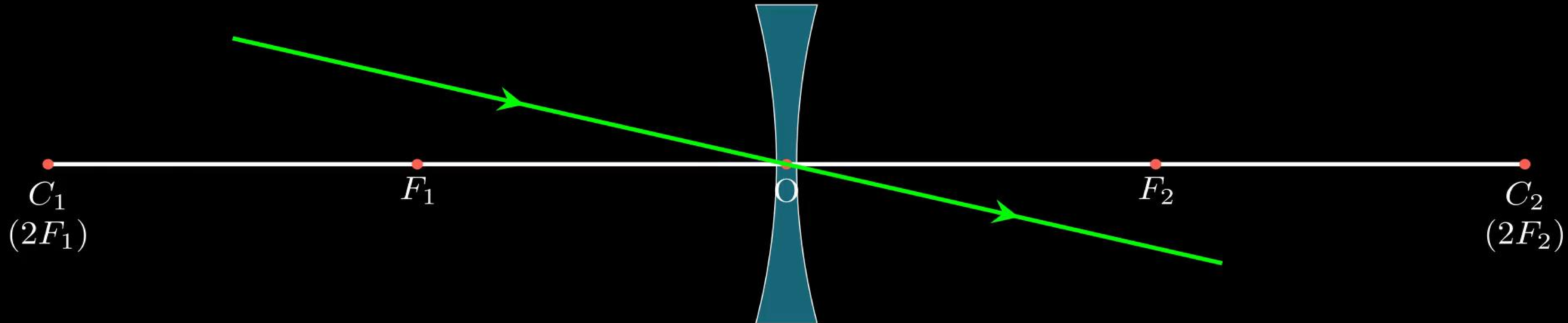
Ray Diagram for Concave Lens

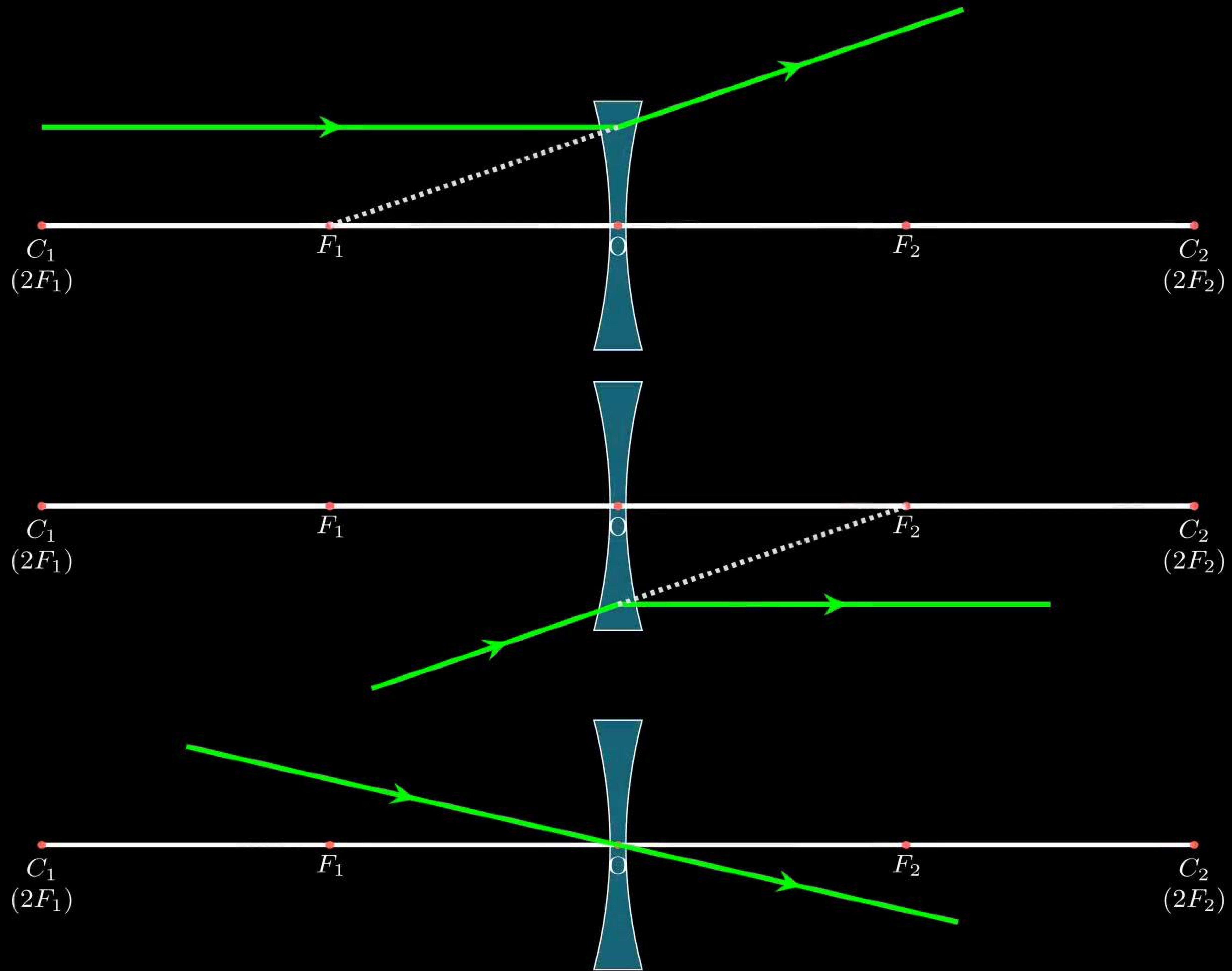
(2) A ray of light appearing to meet at the principal focus of a concave lens, after refraction from a concave lens, will emerge parallel to the principal axis.



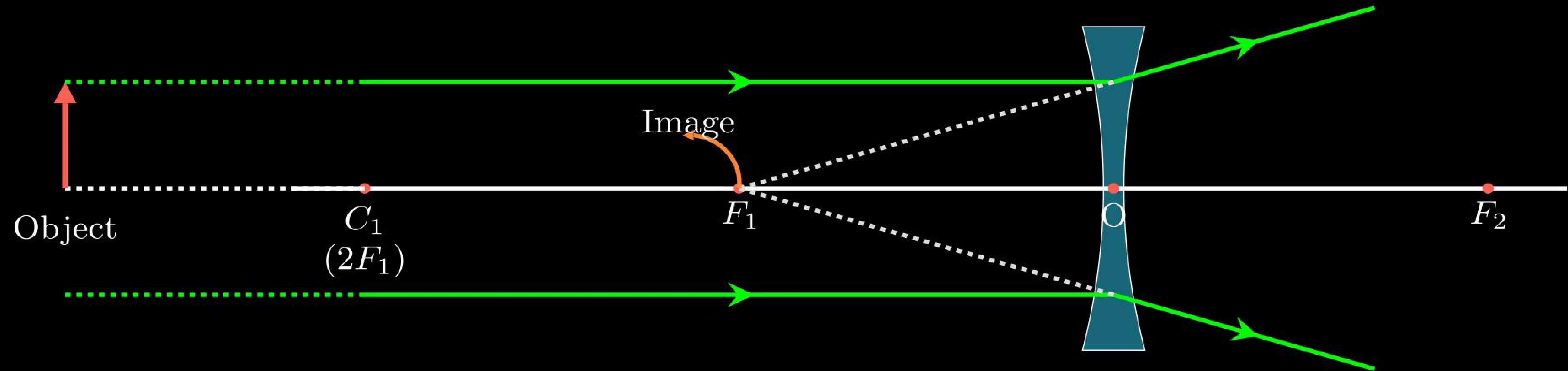
Ray Diagram for Concave Lens

(3) A ray of light passing through the optical centre of a lens will emerge without any deviation.



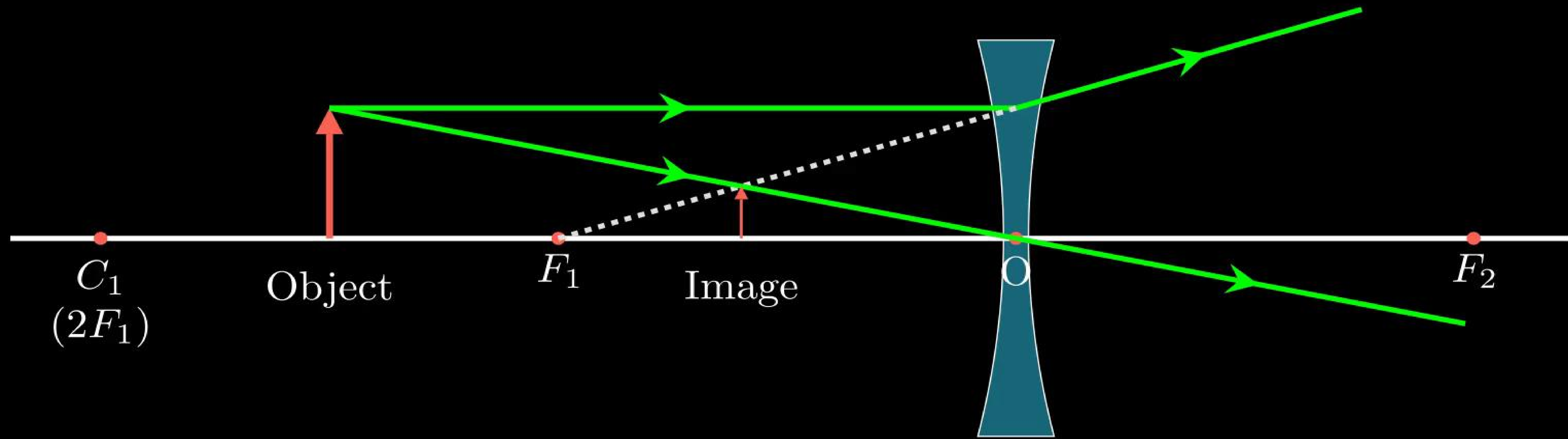


Position of Object (i): At infinity



Position of the Image	Size of the Image	Nature of the Image
At the focus (F_1)	Highly diminished, point-sized	Virtual and Erect

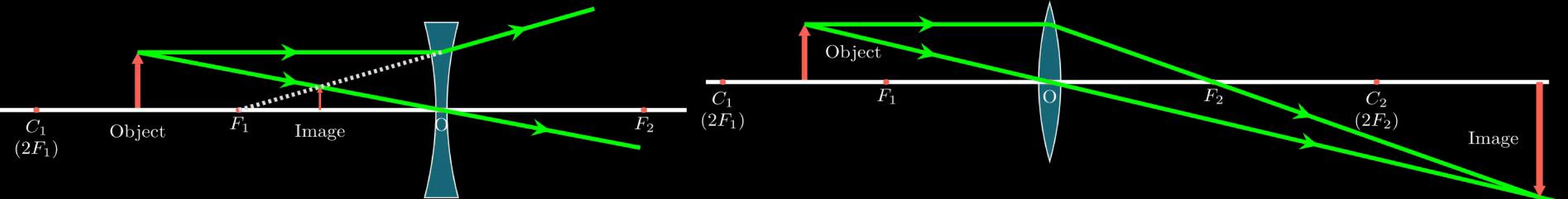
Position of Object (ii): Between infinity and optical centre O of the lens



Position of the Image	Size of the Image	Nature of the Image
Between focus (F_1) and optical centre O	Diminished	Virtual and Erect

Sign convention for spherical lenses :

- For lenses, sign convention are similar to the one used for spherical mirrors. Except that all measurements are taken from the optical centre (O) of the lens.
- Focal length of Convex lens is Positive
- Focal length of Concave lens is negative



Lens Formula and Magnification

- Lens Formula: For spherical lenses, the relationship between u , v and f is given by lens formula.

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

- Magnification (m) : It is the ratio of height of image (h_i) to the height of the object (h_o)

$$m = \frac{h_i}{h_o}$$

- For spherical lenses the magnification m is also related to the object distance (u) and image distance (v).

$$m = \frac{h_i}{h_o} = \frac{v}{u}$$